

DCCREG turbine

Design ledger and fact sheet

The design lock of 2026-10-09, revised the same day after the settlement of its inconsistencies and open points. A temporary documentation ledger and the current fact sheet of the exercise.

status	LOCKED for now — the designer, 2026-10-09: "Let's lock the design for now."
design state	the lock's baseline is commit 09243c7 on branch claude/new-session-0az7f9. The settlement (2026-10-09; §2.3, §6) brought the records into line with it, corrected two computed values (the field at the null with the beads; the beads as drawn) and added first cuts, which stay PROPOSED until the designer accepts them
what the lock means	the design of record below is the baseline; later changes are recorded against it. The lock does not settle what is still PROPOSED or OPEN (§5): those keep their status
this ledger	docs/ledger/DCCREG-design-ledger.md (the source) and .pdf (its print form)
the drawings	docs/ledger/DCCREG-drawings-bundle.pdf: 50 sheets behind a two-page register (§7)
how it is built	python3 docs/ledger/make_ledger.py (needs markdown-it-py, mdit-py-plugins, pypdf and playwright with Chromium)

Abstract. As locked, the DCCREG turbine is a belt-driven tube machine. Its rotor and counter-rotor turn at 600 rpm in opposite directions through a 1 : −1 reversing gear.

- **Two pumps on each shaft half**, both run by the relative rotation at 120 Hz:
 - **the magnetic pump:** three wound utrons pass six iron bridges, wired as the planar dual of a diode charge doubler, and drive the anti-Helmholtz (AH) coil pair;
 - **the electrostatic pump:** an air vane stack in a de Queiroz diode doubler drives two copper rings on the outside of a 50 mm glass vacuum sphere, through two mirrored Cockcroft-Walton chains.
- **Two static fields at the sphere's centre:**
 - the AH's steady magnetic cusp (300 ampere-turns per coil), with its null at the centre;
 - a DC electric field of 7.62 kV/cm (2.57 Pa) at switch-on, settling to 8.2 kV/cm as the insulators leak. It points from ring B to ring A and is steady to 0.19 %.
- **Where the power goes:** the pumps take about 21.5 W from the belt, and the windage and the bearings about 17 W more, about 39 W in all. All of it ends as heat in the clamps, the copper, the iron, the air and the bearings. The products are the fields.
- **What is still open:** the physics is mainstream throughout [OC]. The ratings, the leakage and several parts are placeholders that the bench test must qualify.

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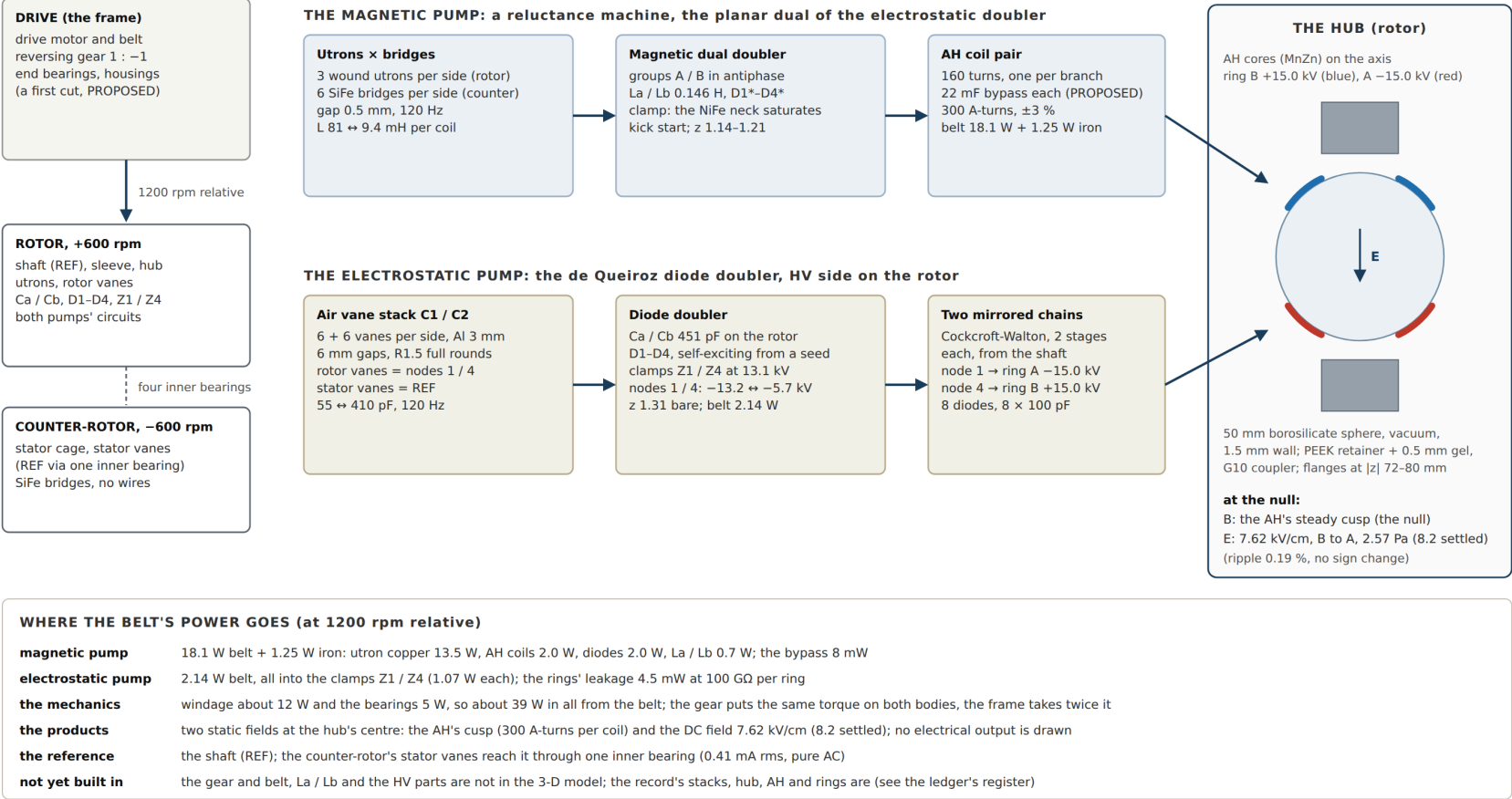
Reading notes.

- **Tags** (`CONVENTIONS.md` §1):
 - [OC] standard, derivable physics;
 - [IR] a modelling or engineering choice, including datasheet-class values;
 - [RH] a heuristic or placeholder, not load-bearing until qualified.
- **Status words** (as the presets use them):
 - DECIDED: by the designer;
 - PROPOSED: worked out here and used, not yet accepted;
 - REGISTER: carried over from an earlier register;
 - DESIGN: the current sizing;
 - OPEN;
 - ESTIMATE.
- **"The record"** is the design of record.
- **Sources:** every number cites the file that holds it; the paths are from the repository root.

1. The machine at a glance

DCCREG turbine, the locked design (2026-10-09): what drives what

Two counter-rotating bodies on one shaft; on each shaft half a magnetic pump and an electrostatic pump; both pumps hold a static field at the hub's centre. Side A below, side B above, mirror images.



Sources: docs/ledger/DCCREG-design-ledger.md · generator: docs/ledger/make_ledger_figures.py

Figure 1. The locked machine: what drives what. The drive turns the two bodies in opposite directions; on each shaft half a magnetic pump and an electrostatic pump run from the relative rotation; both hold a static field at the hub's centre. Source: docs/ledger/figures/architecture.svg (docs/ledger/make_ledger_figures.py).

The bodies.

- **The rotor** carries:
 - the split shaft, which is the electrical reference (REF);
 - the hub;
 - the rotor vanes of the electrostatic stacks;
 - the wound utrons;
 - every circuit part of both pumps.
- **The counter-rotor** carries:
 - the stator cage;
 - the stator vanes, which are REF for the varicaps;
 - the passive iron bridges. It holds no wires, and it reaches the shaft through one inner bearing, the reference link.
- **The frame** holds the two end bearings, the reversing gear and the drive. The gear puts the same torque on both bodies, and its carrier, held by the frame, takes twice it (docs/drive-gear-belt.md §1).
- **The axis** is vertical: side A below, side B above, mirror images about the hub's equator.

The two pumps run at 1200 rpm relative, 120 Hz.

- **The magnetic pump:**
 - each side's three utrons form one group, A or B, whose inductance swings $9.4 \leftrightarrow 81$ mH per coil as the bridges pass;
 - groups A and B run in antiphase, in a circuit that is the exact dual of a diode charge doubler;
 - the saturating NiFe neck in each utron is the clamp;
 - one AH coil sits in each group's branch, and a 22 mF bypass across it (PROPOSED) holds a steady 300 ampere-turns.
- **The electrostatic pump:**
 - the varicaps C1 / C2 swing $55 \leftrightarrow 410$ pF in antiphase;
 - the de Queiroz diode doubler multiplies its charge each cycle until the clamps Z1 / Z4 hold the nodes at their 13.1 kV operating peak;
 - the pump's nodes 1 and 4 then feed two mirrored Cockcroft-Walton chains standing on the shaft, which pump ring A to -15.0 kV and ring B to $+15.0$ kV.

The hub is a 50 mm borosilicate vacuum sphere with a 1.5 mm wall.

- The two MnZn AH cores sit on its axis.
- The two copper-foil rings lie on the glass, under 0.5 mm of silicone gel in a PEEK retainer, with a G10 coupler outside.
- At its centre the AH's null and the rings' field coincide.

The design in numbers

quantity	value	source
speed	600 rpm each way, 1200 rpm relative; both pumps at 120 Hz	sim/pole-design-findings.md §6; sim/core_field.py
tube	940 mm: the air stack of record and the locked hub in solids, 153 solids without a clash	sim/tube-shaft-findings.md §0
diameters	vanes Ø300 mm; bridge ring Ø313 mm; stator cage Ø332 mm	sim/pole-design-findings.md §8
magnetic pump, per side	3 wound utrons (200 turns, 2.23 kg each) and 6 SiFe bridges, gap 0.5 mm	sim/pole-design-findings.md §8
AH	160-turn coils on Fair-Rite 77 MnZn rods (0.80 mm wire, 0.333 Ω as wound); 22 mF bypass each (PROPOSED); 290–308 A-turns (300 mean, ±3 %; 294 as wound)	sim/ah-steady-cusp-findings.md; sim/hub-thermal-findings.md §1

quantity	value	source
the AH's null	at the centre; 0.113 T/m; within ± 0.44 mm over a cycle with the bypass, ± 6.4 mm without	sim/ah-null-findings.md
electrostatic pump, per side	6 + 6 Al vanes 3 mm, full rounds R1.5, 6 mm air gaps; C1 / C2 55–410 pF; Ca / Cb 451 pF	sim/air-stack-sizing-findings.md §6.5
operating peak	13.1 kV at the clamps (the 6 mm gap's breakdown over 1.5)	sim/air_stack_sizing_results.json
rings	Cu foil 0.10 mm, 26.25–55.71° from the axis; beads $\varnothing 3$ / $\varnothing 2$ mm; 29.9 mm apart along the glass	docs/rings-design.md §2
rings' supply	symmetric: -14.96 / $+14.96$ kV on 2 + 2 Cockcroft-Walton stages from the shaft	sim/hub_rings_build_results.json record
field at the null	7.62 kV/cm from B to A at switch-on, 2.57 Pa (the bands alone 7.10); 8.20 kV/cm once settled at 25 °C; ripple 0.014 kV/cm p-p (0.19 %); no sign change in a revolution	sim/hub-beads-settled-findings.md §2; sim/hub_drift_results.json; sim/hub_revolution_results.json
power from the belt	magnetic 18.1 W + 1.25 W iron; electrostatic 2.14 W; windage about 12 W and bearings 5 W	sim/ah-steady-cusp-findings.md; sim/core_field_results.json; sim/rotor-mechanics-findings.md §5
the hub's heat	the AH coils 1.22 W each: the coils at 43 °C, the glass at 32–37 °C in a 25 °C room	sim/hub-thermal-findings.md
cost (placeholders)	the stack of record's build 6,477 EUR; the cheapest qualifying build 6,060 EUR	docs/cost/README.md

2. How the design got here

2.1 What the machine is for, as the record states it

- **The brief now**, in the designer's words (sim/core-null-field-findings.md): the centre of the vacuum core, where the AH null sits, should carry the strongest electrostatic field (pressure).
- **The AH pair** should hold a steady cusp (sim/ah-steady-cusp-findings.md).
- **Both pumps** are the supplies, driven from one belt through the 1 : -1 gear (sim/hub-drive-findings.md).
- **What the field at a magnetic null is to be used for is not documented.** This ledger records the brief as given and does not supply a purpose.

2.2 The path, in eleven steps

Each step asked a question, found an answer in mainstream physics and left a decision behind. The dates are the commits' dates.

#	step (dates)	the question	what was found	what it left
1	the browser tool (to 06-09)	a sectored-disc varicap driving a symmetric Bennet doubler	blocks C-I, M, R, D, T, S in index.html; anchor z 1.203	the tool, now retired and frozen
2	the disc spark-gap machine (06-11 → 06-29)	can a doubler with spark-gap commutation, a resonant tank and islands deliver energy	the doubler's equalisation <i>is</i> the pump; only a downstream sink recovers energy; $\eta \approx 0.70$ is forbidden, $\eta \approx 0.45$ –0.50 (docs/efficiency-resolution.md)	the freeze v0.10 (C_R 789 pF); the efficiency principle
3	the drawn pump re-checked (09-29 → 10-02)	does the designer's drawn pump pump, exactly	exact engines and gates (PUMP-CALC, PUMP-SYNTH, GEOM stage 2, the integrity tool); z 1.325 on the drawing	the tools, frozen
4	the round trip and the design loop (10-03 → 10-05)	do the solids keep the gain	the field-solved floors and strays (about 1 nF) take z to 0.83–0.89 (NO-FLOOR-IN-RANGE); interleaved stacks recover 1.24; with the motor in, z = 1.000	interleaved stacks of vanes are the way to C; the motor cannot sit across Ca
5	the register and the motor (10-05)	can the C-EMs turn the counter-rotor	they are about five orders short, and the pump's own reaction torque (78 mN·m) needs an outside drive	the belt and the reversing gear

#	step (dates)	the question	what was found	what it left
6	the tube (10-05 → 10-07)	an elongated machine with vane stacks along the shaft	C1 1.11 nF, κ 15.7 in vacuum; diodes beat spark gaps (η 1.00, 3.0 W against 1.3 W at η 0.53)	the de Queiroz diode core is the electrostatic pump
7	the pivot: two geared pumps (10-07 → 10-08)	drive the AH from the belt too	the planar dual of the doubler works (z 1.528 against 1.512); wound utrons and passive bridges; the mean-turn fix moves the pick to 2.34 kg, 200 turns, 18.8 W	the magnetic pump of record
8	the air build (10-08)	the first tests run in air	air takes about a third of the vacuum field; 6 mm gaps, full-round vanes, the 6 + 6 cap: 2.14 W; the AH bypass for a steady cusp; the vane matrix and the cost sheet	the electrostatic stack of record
9	the field at the core (10-08)	the strongest field at the AH null	HV side onto the rotor (unchanged pump); cones steady or swinging ($\approx$ 1.4 kV/cm); a DC pair inside the vacuum (65.7 kV/cm, but feedthroughs)	the hub locked; the field needs electrodes
10	the rings (10-09)	rings outside the glass	the beads, not the separation, set the stages; asymmetric 6.93 kV/cm → symmetric 7.10 kV/cm on the bands alone; steady, no swing	the rings and supply of record (docs/rings-design.md)
11	the settlement (10-09)	do the records agree, and what can be settled before the bench	the lock's 18 inconsistencies resolved and 12 more found and settled; with its beads the null reads 7.62 kV/cm; the beads settled, the hub's heat, the AH null, the rotating mechanics, the drive and the parts' first cuts, the model checks (§6)	the decisions of §5.2

2.3 The designer's decisions

date	decision	where recorded
06-16	the freeze v0.10 of the disc machine (12 mm septum, C_R 789 pF)	docs/varcap-design-freeze-v0.10.md
10-03 → 10-05	the design loop's rulings: the machine floats; septum $\leq$ 1000 mm; N = 2; garolite, not PTFE	sim/rt-design-predictions.md
2026-10 (pivot)	rotor and counter-rotor geared 1 : -1; the AH on a magnetic pump; both pumps on the belt	sim/hub-drive-findings.md
10-08	"the first tests run in air"; vanes with full-round edges; the 6 + 6 cap	sim/air-stack-sizing-findings.md
10-08	"the AH pair should hold a steady cusp"	sim/ah-steady-cusp-findings.md
10-08	the reference link through an inner bearing "for now", a brush later	sim/rotor-parts-duty-findings.md §4
10-08	the HV side on the rotor	sim/core-field-findings.md (decision 1)
10-08	the hub locked: layer order, the 50 mm borosilicate sphere, symmetric pumps	presets/hub-locked.json
10-09	the rings outside the glass; a 1.5 mm wall; the AH flanges outside the vessel	presets/hub-locked.json; sim/hub-locked-findings.md
10-09	copper foil with rounded edges for the prototype, a fired-on coating later	sim/hub-rings-build-findings.md (brief)
10-09	"go multistage as long as the rings' separation can hold it"	sim/hub-rings-build-findings.md (brief)
10-09	the symmetric supply: "if it pumps against the core centre, yes!"	docs/rings-design.md
10-09	"Let's lock the design for now."	this ledger
10-09 (the settlement)	keep the stack of record (6 mm gaps, V _{op} = breakdown / 1.5, 22° / 22°, r 150, Ca = 1.1 C _{max}) and the clamps Z1 / Z4	§5.1
10-09 (the settlement)	accept the magnetic pick and the retainer's materials (PEEK, the silicone gel, the G10 coupler outside); the 22 mF bypass and the equatorial split stay PROPOSED	presets/hub-locked.json; §5.1
10-09 (the settlement)	rewrite README.md and CLAUDE.md for the locked machine	README.md, CLAUDE.md

2.4 What was learned, and what it superseded

- **Charge pumps pay their equalisation tax inside the core** [OC].
 - A doubler with lossy transfers cannot recover what it spends, and only a downstream sink recovers energy. The disc machine's claimed $\eta \approx 0.70$ is forbidden (`docs/efficiency-resolution.md`).
 - With smoothly varying capacitance and diodes the transfer itself is lossless ($\eta 1.00$), and that is the pump kept (`sim/diode-stack-findings.md`).
 - Spark gaps, islands, shuttles and the resonant tank are superseded.
- **Strays and floors set the gain, not size** [OC].
 - The disc's field-solved floors and strays scaled with the build (`sim/round-trip-findings.md` §5).
 - Interleaved vane stacks along the shaft give the capacitance without the carriers' penalty, hence the tube.
 - In air the fixed 20 pF node stray is what pulls the capped stack's z from 1.38 to 1.31 (`sim/air-stack-sizing-findings.md` §6.5).
- **A pump's reaction torque must go somewhere** [OC]. The C-EMs could not counter-rotate the stator, so the drive is an outside belt with a 1 : -1 gear, and every reaction torque goes into the frame (`sim/spinup-findings.md`).
- **The dual circuit works.** Inductors for capacitors, currents for voltages and flux for charge give a magnetic pump with its own gain. A saturating neck is its clamp, which makes it the natural driver for the AH coils (`sim/magnetic_doubler.py`, `sim/pole-design-findings.md`).
- **Air sets the voltage, the rims the vanes and the cap the power** (`sim/air-stack-sizing-findings.md`):
 - 6 mm gaps for a 13.1 kV peak;
 - 3 mm full-round vanes for the corona margin;
 - 6 + 6 vanes for 2.14 W.
- **The field at a null needs electrodes; outside the glass they need edges** (`sim/hub-rings-build-findings.md`).
 - The vacuum pair inside reached 65.7 kV/cm but needed two HV feedthroughs.
 - The rings need none. Their beads facing the AH coil ends, not their separation, set how many stages they hold.
- **Solve what is built** [IR]. The beads were sized in local boxes and never put back in the hub, so the record's null missed their 7 %; the drawing's grooves bring the PEEK to the beads, which the boxes had filled with gel (`sim/hub-beads-settled-findings.md`).
- **Only the difference between the rings makes field at the null** [OC].
 - The pump's nodes are twins, so identical circuits give none.
 - The record's mirror chains put the null at the shaft's potential and each ring 15 kV from it, and the field is DC (`sim/hub_revolution.py`).

3. The components and how they work

3.1 Bodies, drive, bearings and the reference

tube-r150-n6-air6-wound-g0p5-6br-hub50: arrangement (g 0.5 / 6 bridges / 1200 rpm), 940 mm long

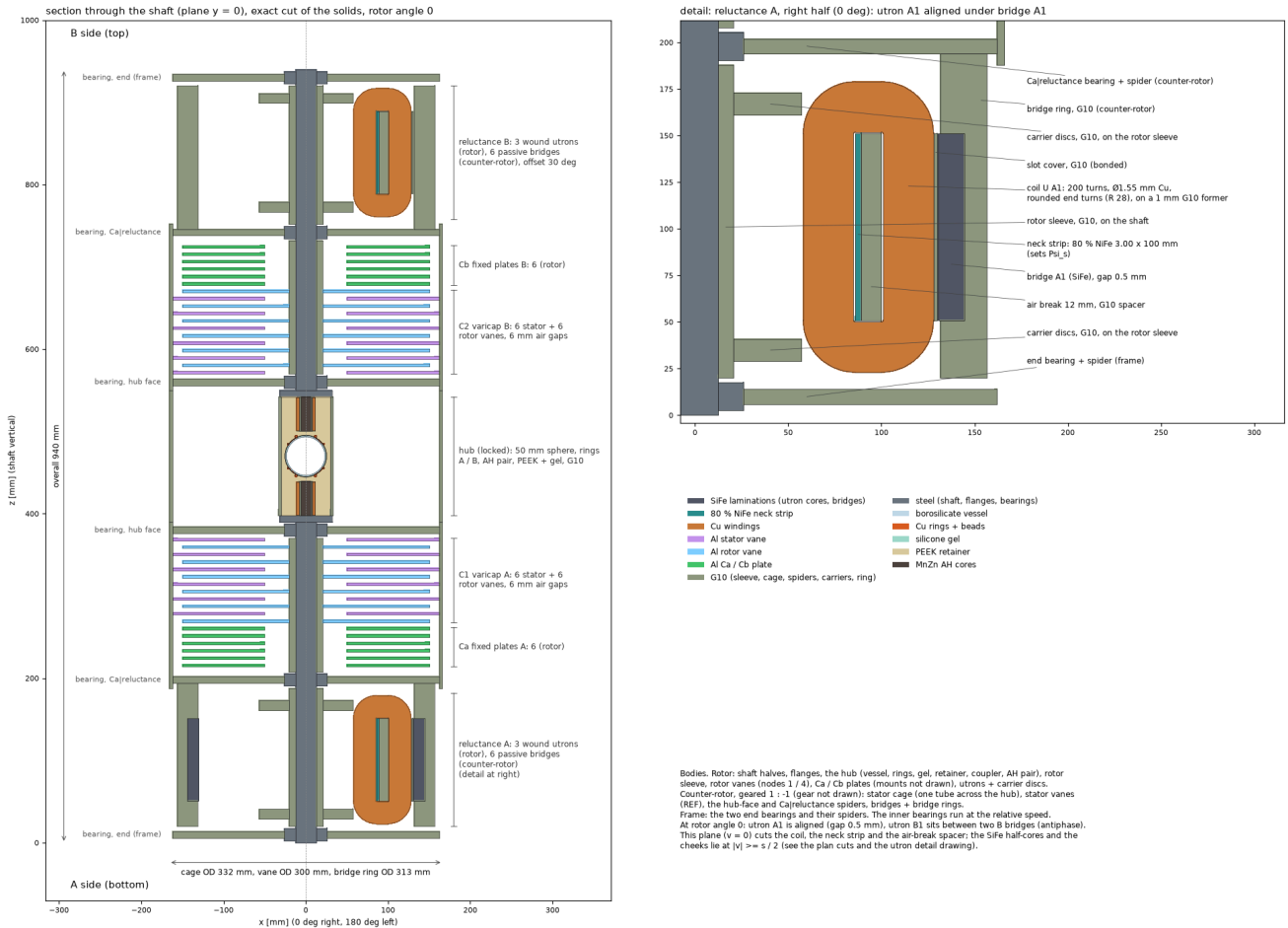


Figure 2. The machine of record in 3-D: the section through the shaft (side A below), with the air stack of record and Ca / Cb on the rotor, the reluctance sections and the locked hub; 940 mm. Source: docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-section.png (sim/tube_geometry.py --record).

- **The rotor** turns at +600 rpm and carries all of both pumps' circuits, so no wire crosses between the bodies (sim/core-field-findings.md §1, §7). It is the split shaft (two halves, each ending in a flange at the hub) with:
 - the G10 sleeve;
 - the rotor vanes (nodes 1 / 4);
 - Ca / Cb, D1-D4 and Z1 / Z4;
 - the utrons on two 12 mm G10 carrier discs per side;
 - La / Lb and D1*-D4*;
 - the hub.
- **The counter-rotor** turns at -600 rpm and carries no HV (sim/tube_geometry.py; sim/pole-design-findings.md §8). It carries:
 - the G10 stator cage (r 162-166 mm), one tube across the hub;
 - the stator vanes (REF);
 - the four inner bearing spiders;
 - six M235-35A bridges per side in a G10 ring, with side B's offset 30°.

- **The frame** holds the two end bearings and their housings, the 1 : −1 reversing gear, the drive motor and the belt. **The drive's first cut** (PROPOSED, docs/drive-gear-belt.md):
 - a bevel reverser at side A, its carrier held by the frame: two stainless side gears (z 30, m 2) and three POM-C pinions;
 - an HTD 5M belt, 4 : 1, from a 200 W brushless motor whose driver ramps and limits the current;
 - the gear puts 0.48 / 0.52 N·m on the bodies and 1.00 N·m into the frame; the belt carries 59 W with a 25 W windage allowance; the run-up takes 32 s;
 - the pumps' 120 / 240 Hz pulsation stays in the bodies: the gear sees 0.061 N·m p-p;
 - 1,066 EUR against the cost sheet's 800.
- **The bearings:** six 6205-class deep-groove bearings (25 / 52 / 15 mm), each on an 8 mm G10 spider.
 - The four inner ones join rotor and counter-rotor at the relative speed; only the two end bearings face the frame.
 - **So the counter-rotor floats on the shaft** (sim/tube-shaft-findings.md §0; sim/rotor-mechanics-findings.md §4, an independent model that agrees). As laid out (d 25) the first mode is 27–29 Hz, and 1 g lateral moves the utron gap 123 μm at the stack's centre and 1.7× that at its outer end.
 - **The ways out:** d 40 gives 65 Hz and 25 μm. A bearing between the shaft and each bridge ring's outer end gives 131–165 Hz (realistic to stiff seats) and 2–3 μm, but on G10 spiders it sits only 9 % above the pumps' 120 Hz, and the layout grows 10–20 mm a side for it. The choice is the designer's (§5.2).
 - **The gap budget:** each inner bearing's total runout within 0.027 mm as laid out (P5 bearings under an axial spring preload), or 0.05 mm with the eighth pair (sim/rotor-mechanics-findings.md §5).
- **The shaft:**
 - it is d 25 in the tube model and d 30 in the hub's register: OPEN, with the bearings (§5.2). A 6205 needs 25 mm journals;
 - non-magnetic, $\mu_r \leq 1.05$: annealed austenitic stainless. Steel halves would tie the AH null to how each seats, and titanium drops the first mode to 21 Hz (sim/ah-null-findings.md §5; sim/rotor-mechanics-findings.md §4);
 - it is split at the hub, each half bolted at its flange (r 32 × 8 mm at ±72–80 mm in the locked hub).
- **The rotating loads** (sim/rotor-mechanics-findings.md):
 - no banding is needed: at 750 rpm every part holds with a factor of 19 or more, if the studs are preloaded and the joints bonded in the impregnation;
 - balance each body to G2.5;
 - windage takes about 12 W and the bearings 5 W.
- **The reference link.** The counter-rotor's stator vanes are REF for C1 / C2. They reach the shaft through one inner bearing:
 - the path is rail → lead → outer ring → balls → inner ring → shaft;
 - it carries the varicaps' displacement current, 0.41 mA rms of pure AC with no DC (sim/core_field_results.json).
 - A floating counter-rotor would need about 10 nF to the shaft, and at 100 pF the pump stops, so the bearing link stays. It is not acceptable with a spark dump or without clamps, where the surges need a brush (sim/rotor-parts-duty-findings.md §4).

3.2 The magnetic pump: utrons, bridges and the dual doubler

How a bridge makes an inductance that swings [OC].

- A utron is a U-core electromagnet on the rotor. Its two radial tips face the counter-rotor at r 130 mm.

- When a passive SiFe bridge spans both tips, the flux closes across two 0.5 mm gaps and the coil's inductance is high: 81.0 mH. Between bridges the nearest iron is 9.74 mm away and it falls to 9.4 mH (κ 8.6).
- Six bridges on a 60° pitch at 1200 rpm relative swing it at 120 Hz. Each bridge sees the three utrons pass, so it carries unipolar flux at 60 Hz ([sim/pole-design-findings.md](#) §8).

The dual doubler ([sim/magnetic_doubler.py](#)).

- The electrostatic diode doubler's circuit graph has a planar dual: capacitors become inductors, voltages currents, charge flux.
- In it, the two varicaps become the two utron groups $L1(\theta) / L2(\theta)$, three utrons in series each (0.243 H). Ca / Cb become the chokes La / Lb (0.146 H), and the diodes become D1*-D4*.
- The bridges of side B sit half a pitch on from side A's, so the groups are in antiphase.
- When a group's inductance falls while its flux is held, its current rises: it generates, and the belt does the work against the bridges' pull. When the inductance rises the group motors. The groups swap every half cycle.
- Per cycle the stored flux grows by a fixed factor: z 1.21 in the design screen's linear model, against a selection rule of ≥ 1.20 . The loaded transient run reads 1.139 early on, and 1.147 with the AH bypass ([sim/pole_design_variants_op.json](#); [sim/ah-steady-cusp-findings.md](#)).
- Its duality check against the electrostatic doubler gave z 1.528 against 1.512 ([sim/hub-drive-findings.md](#)).

Wound utron, detail: 200-turn yoke coil on a split U-core with a NiFe neck, against a passive bridge (all dimensions in mm)

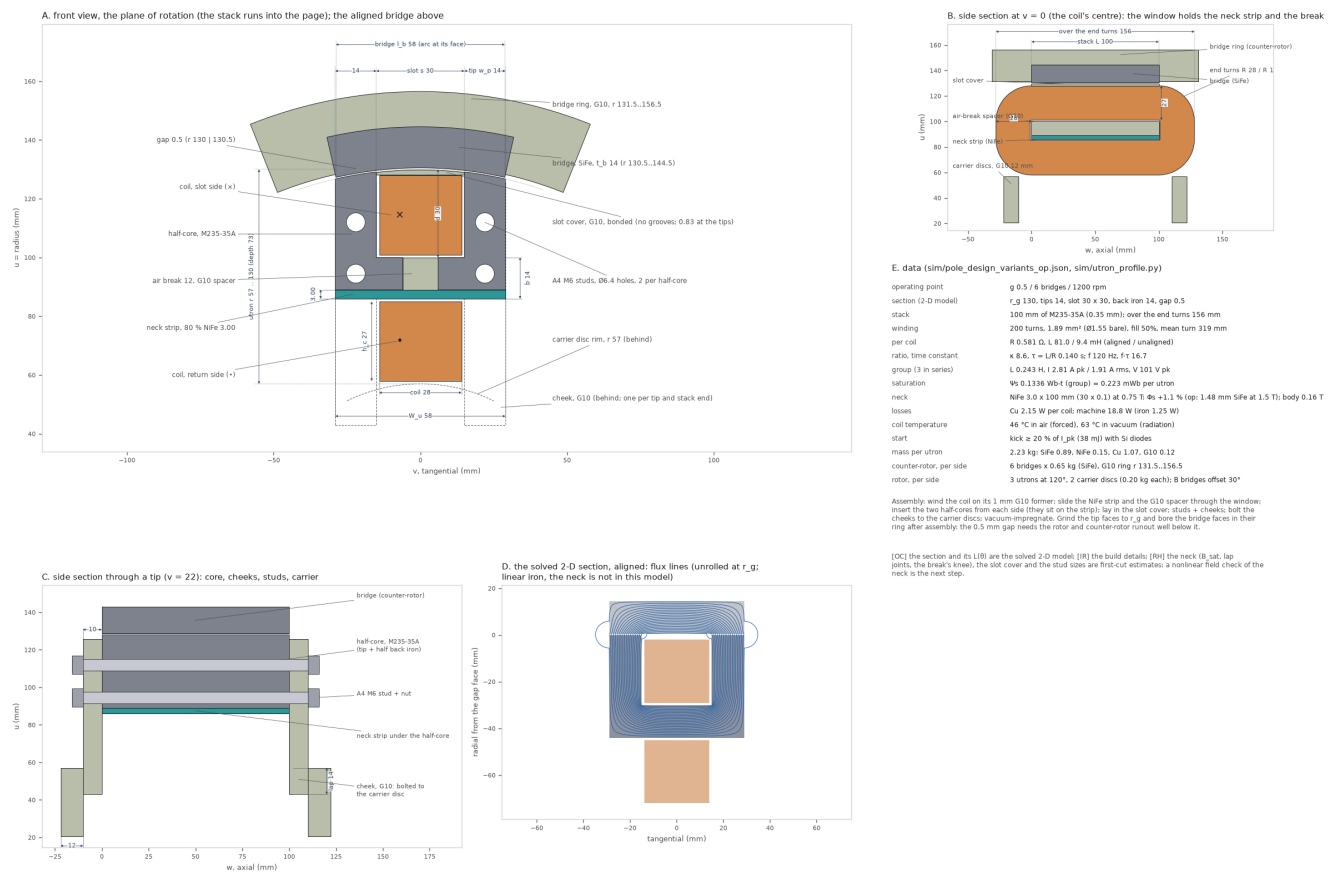
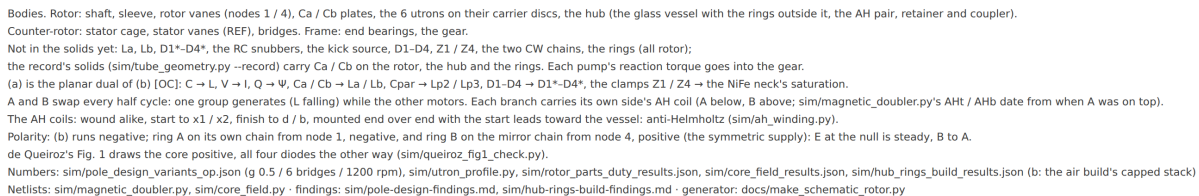


Figure 3. The wound utron against a bridge: the 200-turn coil on the split U-core, the 3.0 mm NiFe neck, the bonded slot cover, the studs; the solved 2-D flux. Source: [docs/figures/utron-core-detail.png](#) ([docs/make_utron_drawing.py](#)). The half-core's manufacturing drawing is DCCREG-UTR-101 (bundle sheet 4).

The clamp, the start and the parts.

- **The clamp.** The dual of the avalanche clamps is the iron's saturation. A 3.0 mm 80 % NiFe strip under each utron's back iron saturates at Ψ_s 0.134 Wb-turns per group, and the current then stops growing. The law $i = \Psi/L \cdot (1 + (\Psi/\Psi_s)^6)$ is a fit [IR]. The neck is sized to the AH's need and no larger, because heat goes as Ψ_s^2 (sim/pole-design-findings.md §3, §8).
- **The start.** With silicon diodes (0.55 V) a small seed does not grow, because the winding voltage while growing is only a few volts. A one-time kick of about 20 % of Ψ_s starts it: it seeds 0.110 A and 3.5 mJ (sim/pole-design-findings.md §7). Kick above about 200 rpm, clear of the drive's 20 Hz gear mode (docs/drive-gear-belt.md §4.2). The kick source is OPEN, and so is its polarity if the bypass electrolytics are polarised.
- **The parts:**
 - the utrons: M235-35A 0.35 mm laminations, stack 100 mm, 200 turns of Ø1.55 mm Cu, 0.58 Ω , 2.23 kg each as built;
 - La / Lb: 0.146 H DC chokes, 0.87–1.15 A, ≤ 110 V; first cut a gapped EI, "not designed" [RH];
 - D1*–D4*: silicon, 2.8 A peak, reverse 112 / 101 / 43 / 45 V; parts not chosen;
 - snubbers: an RC at each of the eight nodes (21 nF + 243 Ω) [IR];
 - the wiring stray Lp2 / Lp3: 4.4 mH [IR];
 - the coils run at 46 °C in air (63 °C in vacuum), at the model's 40 °C ambient (sim/pole_design.py).
- **The power:** 17.6 W on the belt without the bypass (utron copper 12.9, AH 2.00, La / Lb 0.66, diodes 2.01), plus 1.25 W of iron. With the bypass the belt reads 18.1 W (docs/schematic-rotor-circuits.png; sim/ah-steady-cusp-findings.md).

(a) the reluctance pump drives the AH pair · (b) the electrostatic pump's rings hold a DC field on the AH null; only C1 / C2 straddle the bodies · operating point g 0.5 / 6 bridges / 1200 rpm relative · only diodes switch



3.3 The AH pair and the steady cusp

- 12 / 33

- **The null's position** (`sim/ah-null-findings.md`):
 - balanced, it sits at the centre, with 0.113 T/m along the axis (0.377 mT/m per A-turn on both coils; 0.111 T/m as wound), $-\frac{1}{2}$ of that across. The rods raise it 4.8×;
 - each A-turn of imbalance moves it 25 μm toward the weaker coil. With the bypass it stays within ± 0.44 mm over the 120 Hz cycle; without it it swings ± 6.4 mm, beyond the ± 5 mm where the rings' field holds 4.4 %;
 - the earth's vertical field moves it 0.35–0.5 mm; a 0.5 mm misplaced coil, 0.25 mm.
- **The steady cusp.**
 - Without a bypass, each coil carries its branch's unipolar 139–449 A-turns at 120 Hz. Top and bottom peak in turn, so the null moves along the axis every cycle.
 - A 22 mF capacitor across each coil (34 Hz with the coil, far below 120 Hz) carries the ripple. Each coil then holds 290–308 A-turns (300 mean, ± 3 %), and top and bottom stay within 17 A-turns (`sim/ah-steady-cusp-findings.md`).
 - 2.2 mF would resonate near 107 Hz and make the ripple worse. The capacitor's duty is light: 0.65 A rms, about 0.5 V, 8 mW for the pair.
- **The cost of steadiness:** the steady field is the mean, 300 A-turns, against the 449 A-turn peak the AH was sized to. A steady 450 needs 240 turns or a larger pump, which is not re-sized.
- **The rods** peak at 0.121 T at 300 A-turns. They reach the register's 0.30 T limit [RH] at 743 A-turns for the AH alone; the record's "about 600" scaled a run with the cone windings in series (`sim/ah-null-findings.md §4`).

3.4 The electrostatic pump: the vane stack and the diode doubler

Electrostatic vane stack for AIR: 6 mm gaps, 6 sectors of 22° (stator) / 22° (rotor), 6 + 6 vanes per varicap per side; vanes and Ca / Cb plates Al 3 mm with full-round edges R1.5 (dimensions in mm)

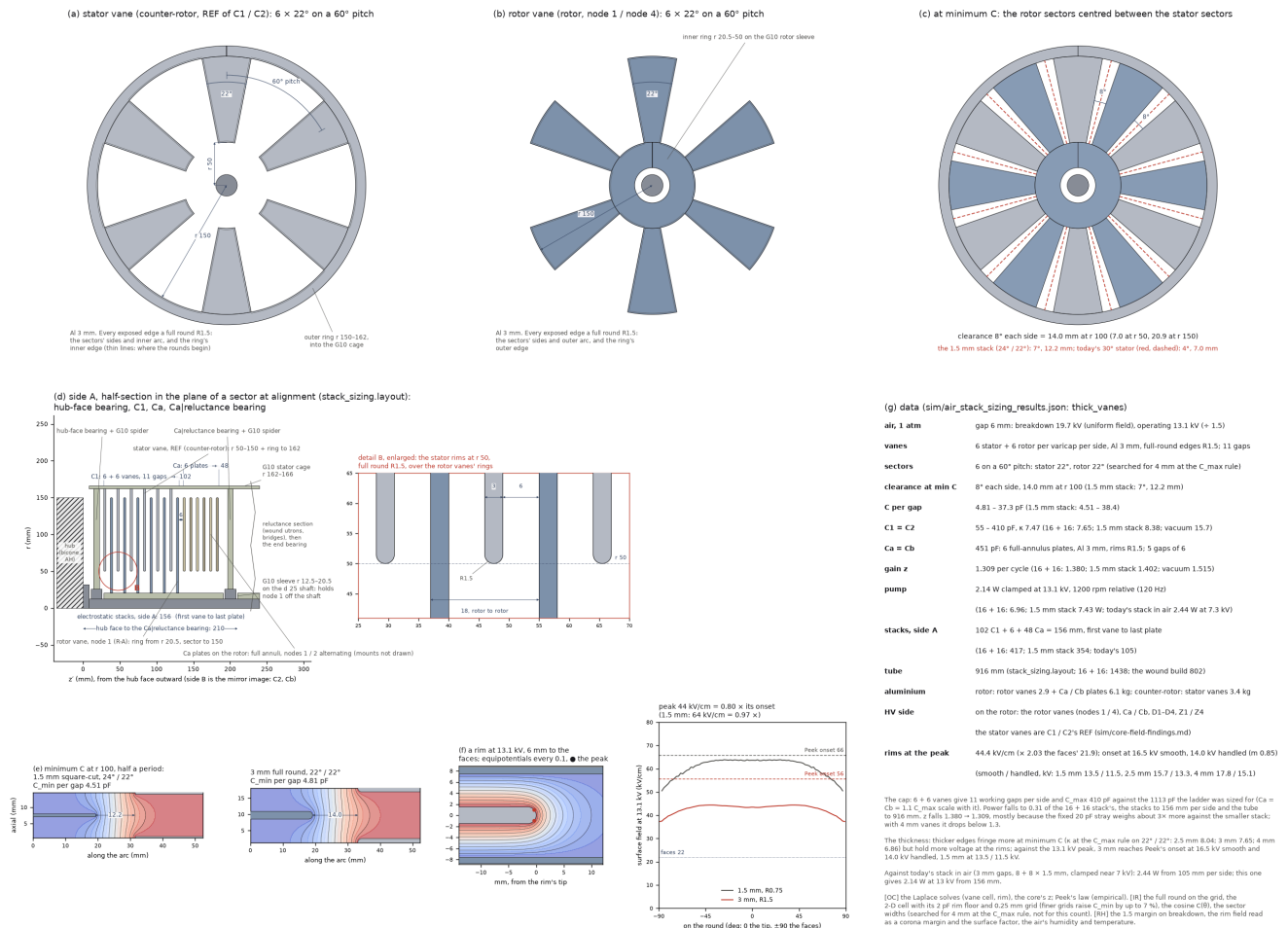


Figure 5. The air vane stack of record: the stator vane (REF), the rotor vane (node 1 / 4), both at minimum capacitance, side A's half-section, the vane cell and the rim field against Peek's onset. Panel (d) still draws the placeholder hub. Source: `docs/figures/air-vane-stack-6mm.png` (`docs/make_air_vane_drawing.py`).

The varicaps C1 / C2 (`sim/air-stack-sizing-findings.md §6.5`).

- On each side, 6 stator vanes (REF, counter-rotor) interleave with 6 rotor vanes (node 1 on side A, node 4 on side B) across 11 gaps of 6 mm in air.
- The vanes are aluminium, 3 mm thick, every exposed edge a full round R1.5. Each has six 22° sectors on a 60° pitch, r 50–150 mm.
- With the sectors aligned the stack is 410 pF. With the rotor's sectors centred between the stator's it is 55 pF (κ 7.47).
- C1 and C2 sit half a pitch apart, so they swing in antiphase six times a relative revolution: 120 Hz.

The doubler [OC] (`sim/core_field.py`; de Queiroz's symmetrical generator, its Fig. 1 with every diode reversed: the same circuit at negative polarity).

- **The circuit:**
 - C1 from node 1 and C2 from node 4 to REF;
 - transfer capacitors Ca (1-2) and Cb (3-4) of 451 pF;
 - four diodes: D1 2→0, D2 3→0, D3 1→3, D4 4→2.
- **One half cycle, step by step:**
 1. The capacitor that is shrinking keeps its charge, because its diode is blocked, so its voltage rises as Q/C . The rotation does work against the vanes' attraction.
 2. Through Ca or Cb that lift pulls the next node until a diode conducts.
 3. The charge then pours onto the other varicap while it is large, so it accepts the charge at low voltage.
- **The gain.** Charge is collected while a capacitor is large and lifted while it is small, so it grows each cycle by a fixed factor: z 1.31 bare (1.3095 in ngspice). With the rings' chains attached, z is 1.191 at start-up.
- **Self-excitation.** Between diode events the circuit is linear and scale-free, so any seed in the growing mode (a contact potential, a triboelectric charge) multiplies by z every cycle [OC].

The clamps and the operating point.

- $z > 1$ at every amplitude, so something must stop the growth. The clamps Z1 (node 1) and Z4 (node 4) are avalanche strings, first cut 66×200 V. They break down at the operating peak V_{op} 13.13 kV, conduct once a cycle near the peak and turn the whole surplus into heat: 1.07 W each, 0.95 mA peak (`sim/core_field_results.json`).
- **V_{op}** is the 6 mm air gap's breakdown, 19.7 kV, over a 1.5 margin [RH].
- **The rims' field** at 13.1 kV is 44.4 kV/cm, 0.80 of Peek's onset (55.7 kV/cm). Corona starts at 16.5 kV on a smooth surface and 14.0 kV on a handled one, margins of $\times 1.26$ and $\times 1.07$ (`sim/air_stack_sizing_results.json`).
- **The nodes:**
 - nodes 1 and 4 swing $-13.2 \leftrightarrow -5.7$ kV, half a cycle apart (mean -8.6 kV);
 - nodes 2 and 3 swing $0 \leftrightarrow -6.1$ kV.
 - D1–D4 see reverse peaks of 6.1 / 6.1 / 13.2 / 13.2 kV (parts not chosen; the guidance is $\times 2$ rated HV sticks with a surge resistor).
- **The power:** the belt pays 2.14 W, all of it into the clamps.
- **Without clamps** the pump runs as a relaxation oscillator: it arcs at the stack, sends 25–47 A surges through the diodes and 38 A through the bearing link, so the clamps stay (`sim/electrostatic-no-clamp-findings.md`).
- **Ca / Cb:** 1.1 C_{max}, six full-annulus aluminium plates per side with 5 gaps of 6 mm, on the rotor. Their mounts need 5.7–7.5 kV of creepage and are not drawn.
- **Strays:** 20 pF per node [RH]. Each pF costs about 0.003 of z (`sim/core-field-findings.md §4`).

3.5 The hub

Source: `presets/hub-locked.json` (locked 2026-10-08; the decisions of 2026-10-09).

layer, inside out	what	status
the vessel	borosilicate sphere, OD 50.0 mm, wall 1.5 mm (R_{in} 23.5 mm), ϵ 4.6, vacuum inside; about 0.8 MPa shell stress	DECIDED
the rings	two copper-foil bands on the outer surface (§3.6)	DECIDED / DESIGN
the AH	the MnZn rods with their 160-turn coils on the axis, PEEK seats at ± 25 –30.7 mm (§3.3)	DECIDED (placement) / REGISTER (dimensions) / DESIGN (the winding) / PROPOSED (the ends rounded to 1.5 mm)
the retainer	unfilled PEEK, ϵ 3.2, $r \leq 30$ mm, ± 72 mm; a 0.5 mm pocket over the glass with grooves for the beads	DECIDED (role; material, 10-09) / PROPOSED (the grooves' full-round tops; the dimensions' first cut)
the interface filler	two-part silicone gel, degassed and vacuum-cast into the pocket, ϵ 2.9, 0.5 mm	DECIDED (10-09)
the shaft coupler	G10 outside the PEEK, r 30–33 mm, ϵ 4.7	DECIDED (role; G10, 10-09) / PROPOSED (a straight tube over the retainer and both flanges)
the shaft	split, flanges r 32 $\times$ 8 mm at ± 72 –80 mm; non-magnetic	DECIDED (layout) / REGISTER (d 30) / OPEN (with the bearings)
the pumps	each shaft half carries its side's electrostatic stack and utron group, symmetric top and bottom	DECIDED

- **Why PEEK and gel:**

- the glass–retainer interface is what holds the DC between the rings, and a dry machined fit leaves air gaps; the gel fills them, beds the beads and takes up the PEEK–glass expansion mismatch;
- PEEK is non-magnetic and non-conducting, machines well and barely moves the field (-1.5% to $+0.7\%$ from alumina to PTFE);
- it is the ten-times-less-conductive PEEK, against the glass and gel, that sets the drift (§3.7) (`sim/hub-rings-build-findings.md` §1).

- **Once the DC has settled** (minutes to hours; `sim/hub-beads-settled-findings.md`), conduction shares it by the conductivities [OC]:

- the beads relax (0.69 / 1.88 kV/mm in the gel), and the PEEK, ten times less conductive than the gel, takes the DC: 2.3 kV/mm across the AH seat on average;
- the drawn grooves' square top corners then hold 7 kV/mm in the PEEK and the AH end's square corner 7.9. Proposed: the grooves' tops full-round with 1.0 mm of gel (4.1 kV/mm) and the AH ends rounded to 1.5 mm at REF (4.5 kV/mm), about a fifth of PEEK's short-term strength. The AH seat stays 5.7 mm;
- **the equatorial beads set a condition:** their gel holds 5 kV/mm only while the glass conducts at most 4.1 times the gel (4.4 with the full-round grooves). With the glass at 40 °C ($5.4\times$) they reach 6.15 kV/mm;
- the bead–glass contact wedge stays finite, and the voltage across it at most half of air's Paschen breakdown.

- **The hub's temperature** (`sim/hub-thermal-findings.md`): the AH coils dissipate 1.22 W each. In a 25 °C room, with the air between the vane stacks 5 K above it [RH], the coils run at 43 °C and the glass at 32–37 °C, 32.5 °C between the rings.

- **The vessel's resistivity**, $1e13 \Omega\cdot m$ at 25 °C [IR], falls with 0.90 eV (the record's two points). Between the rings in service it is $4.2e12 \Omega\cdot m$, 2.4 times the gel's conductivity: the beads hold, up to a room of about 30 °C if the gel's conductivity does not rise with temperature. The bench measures both (phase 3).

- **The coupler and the leads** (first cut, PROPOSED; `docs/rings-design.md` §2):

- a straight G10 tube, $\varnothing 60$ / $\varnothing 66$ mm over $|z| \leq 80$ mm, holding the PEEK retainer and both shaft flanges (turned to $\varnothing 60$), each bonded and pinned;
- each ring's lead leaves its equatorial bead along the bead's normal, through the PEEK and the coupler, and runs over the coupler to its own end;
- the air outside sees at most 0.55 kV/mm.

3.6 The rings and their symmetric supply

The rings (docs/rings-design.md; drawing DCCREG-HUB-201, Figure 6).

- **The bands.** Two bands of annealed Cu-ETP foil, 0.10 mm, lie on the vessel's outer surface from 26.25° to 55.71° from the axis. Ring B is the upper one, ring A its mirror.
 - Each band is 12.85 mm along the glass and 13.1 cm², and the equatorial gap is 29.9 mm along the glass.
 - Each band is cut as 12 gores, 5.79 → 10.81 mm wide, overlapped 1 mm and soldered, and burnished on. There is no adhesive film: the gel bonds it.
- **The edges are beaded.** A bare foil edge concentrates the field, so each edge carries a soldered copper wire ring:
 - Ø3 mm at the polar edge, 73.6 mm of wire;
 - Ø2 mm at the equatorial edge, 135.0 mm of wire;
 - each bedded in a groove of the PEEK, 3.5 / 2.5 mm deep (proposed: full-round tops with 1.0 mm of gel, 4 / 3 mm deep).
- **Why these angles.**
 - The polar bead faces the AH coil's end across 5.4 mm of PEEK. A bigger bead comes closer to it, so the polar edge does not ease with size, and a ring at 15 kV needs its polar edge at 26° or more.
 - The gap across the equator holds the DC at 1 kV/mm along the glass.
 - Both beads hold 5 kV/mm in the gel [RH]: 4.98 / 4.46 kV/mm as drawn (4.85 / 4.36 in the build's box, which had gel throughout). The polar bead sits 0.4 % under its rating.
 - The rings' separation is not the limit; the beads are (sim/hub-rings-build-findings.md §2, §4).
- **Later:** a fired-on coating needs the same beads, or a resistive grading toward the pole.

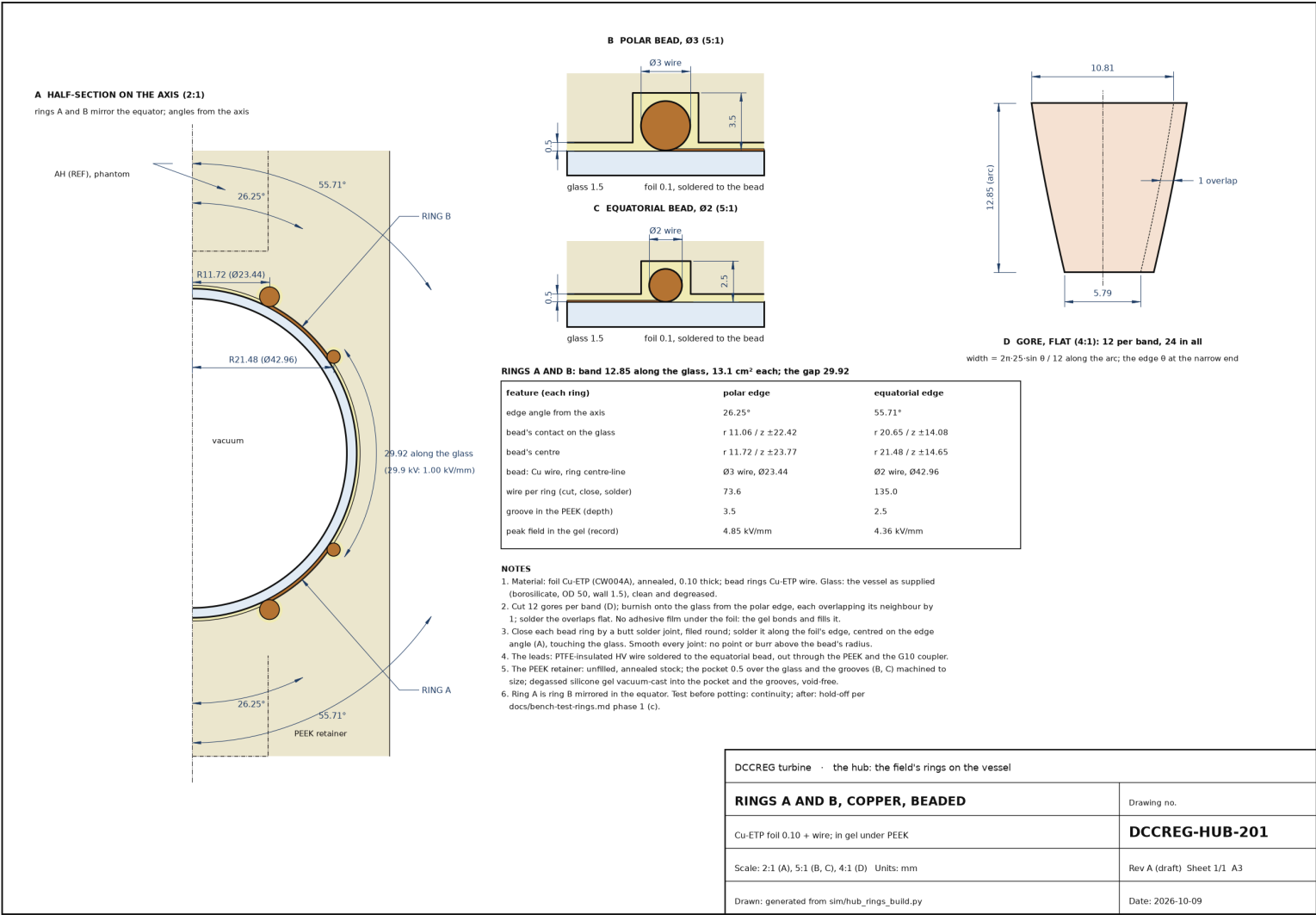
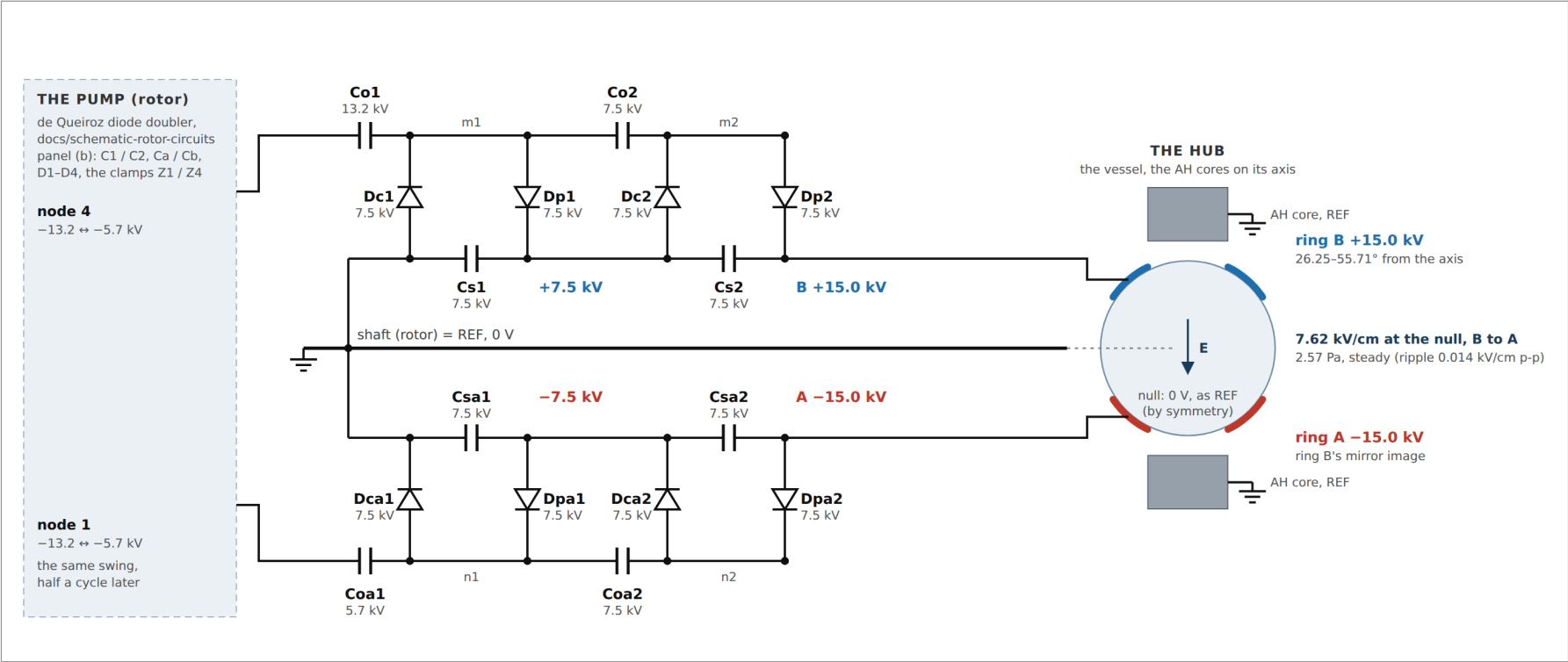


Figure 6. DCCREG-HUB-201, the rings' manufacturing drawing: the half-section at 2:1, the beads in their grooves at 5:1, one gore flat at 4:1, the feature table and the build notes. Source: docs/drawings/DCCREG-HUB-201.pdf (docs/make_rings_drawing.py).

DCCREG turbine: the rings' DC supply of record, the symmetric pair (each ring on its own chain from the shaft)

Ring B on 2 positive Cockcroft-Walton stages driven by the pump's node 4, ring A on the mirror image driven by node 1; both chains stand on the shaft (REF), which the AH and the null share. 100 pF storage, 100 GΩ per ring.



chains	2 stages each; ring B's from node 4 (Co, Dc, Dp, Cs), ring A's from node 1 (Coa, Dca, Dpa, Csa), the diodes reversed
levels	each stage adds node 4's / node 1's swing: 7.5 kV a stage at 100 GΩ per ring; 29.9 kV across the rings
diodes	8 HV stacks, reverse ≤ 7.5 kV each (20 kV stacks, derated 1.5×)
capacitors	8 × 100 pF / 30 kV; each holds about one stage's 7.5 kV DC, but Co1 13.2 and Coa1 5.7 kV: nodes 4 and 1 sit at -8.6 kV mean
ripple	ring A 34 V, ring B 29 V p-p at 120 Hz; ring A tops up as node 1 bottoms, ring B as node 4 peaks, 0.124 of a cycle later: 57 V p-p across the gap, 0.2 % of the field; no swing
pump	belt 2.14 W into the clamps; leakage 4.5 mW; start-up gain z 1.191
rings	Cu foil bands, beads Ø3 / Ø2 mm (polar / equatorial) in gel; strays 5.1 pF each to REF, 1.1 pF between

Numbers: sim/hub_rings_build_results.json (record, record_supply), sim/hub_drift_results.json (swing), sim/core_field.py (the deck) · findings: sim/hub-rings-build-findings.md · generator: docs/make_rings_supply_schematic.py

Figure 7. The rings' DC supply of record: ring B on two positive Cockcroft-Walton stages from node 4, ring A on the mirror image from node 1; both chains stand on the shaft; each capacitor's DC and each diode's reverse peak. Source: docs/schematic-rings-supply.svg (docs/make_rings_supply_schematic.py).

The supply: two mirrored Cockcroft-Walton chains [OC] (Figure 7; `sim/core_field.py` `dc, n_cw = n_cw_a = 2, a_ref = "shaft"`).

- **Ring B's chain:**
 - node 4 → Co1 → m1 → Co2 → m2;
 - clamp diodes Dc1 (shaft → m1) and Dc2 (b1 → m2), peak diodes Dp1 (m1 → b1) and Dp2 (m2 → ring B);
 - smoothing capacitors Cs1 (shaft-b1) and Cs2 (b1-ring B).
 - Each stage lifts the smoothing column by node 4's swing, 7.5 kV.
- **Ring A's chain** is the same ladder on node 1 with every diode reversed (Coa, Dca, Dpa, Csa). It lowers its column by 7.5 kV a stage.
- **How it pumps against the core centre.**
 - Both chains stand on the shaft (0 V). The hub is mirror-symmetric and ring A is ring B's negative, so the null — and the whole equatorial plane — sits at the shaft's potential with nothing connected to it [OC].
 - Each ring stands 15 kV from it, as far as its polar bead allows.
- **The parts and levels:**
 - 8 HV diode stacks, all at 7.5 kV reverse;
 - 8 capacitors of 100 pF / 30 kV, each holding a stage's 7.5 kV DC, except Co1 at 13.2 kV and Coa1 at 5.7 kV (nodes 4 and 1 sit at −8.6 kV mean);
 - the levels: b1 +7.49 kV, ring B +14.96 kV, a1 −7.49 kV, ring A −14.96 kV; about 27 mJ stored;
 - no Dk and no C_A (`sim/hub_rings_build_results.json` record_supply).
- **Why 100 pF:** it keeps the pump's start-up gain at 1.191 (1.03 at 1 nF), and still smooths the ripple to tens of volts.
- **The strays:** 5.1 pF from each ring to REF and 1.1 pF between the rings, behind diodes.

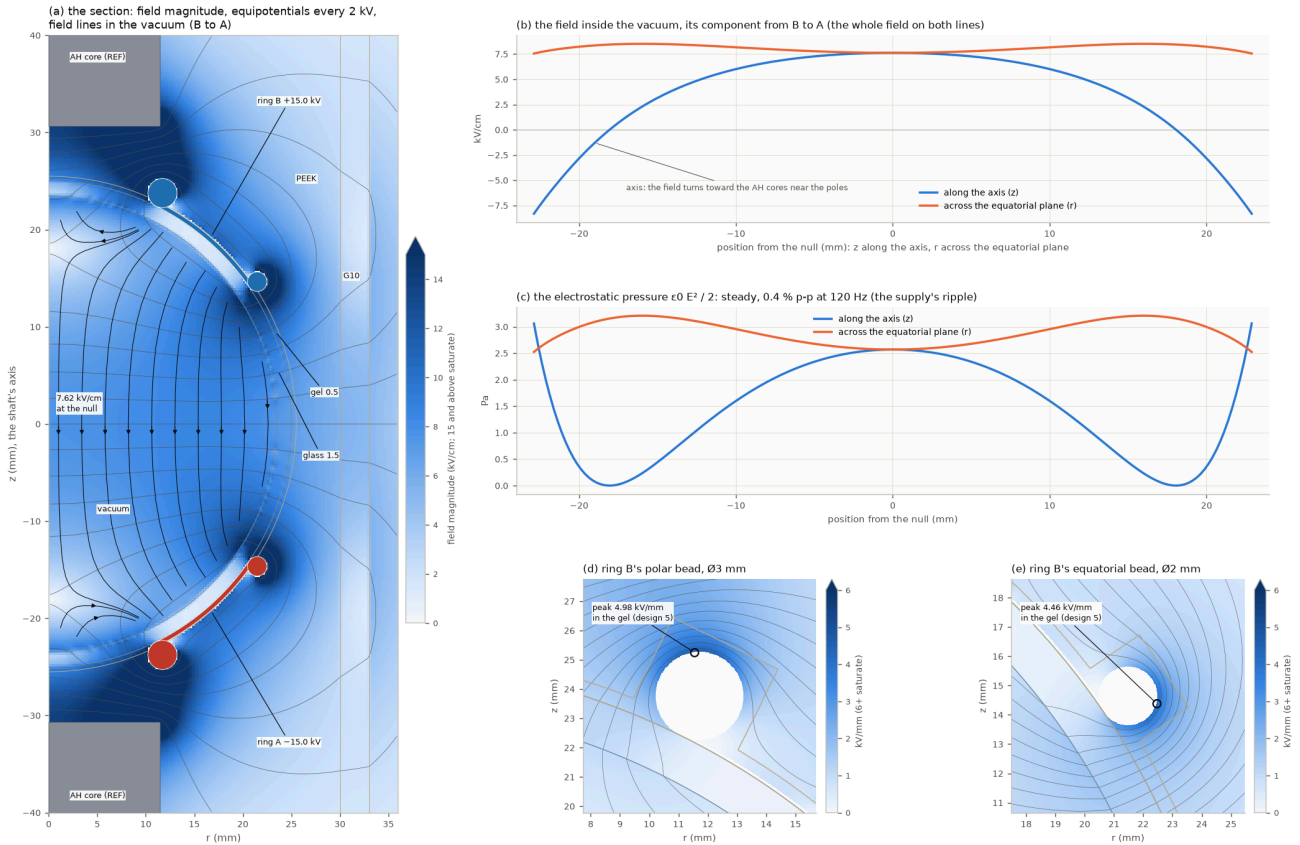
3.7 The field at the null, in space and in time

How the field is computed [IR] (`sim/core_rings.py`, `sim/hub_locked.py`, `sim/hub_rings_build.py`).

- An axisymmetric finite-volume solve of $\text{div}(\epsilon \text{ grad } V) = 0$ on the half hub, with 0.25 mm cells.
- REF is the AH envelope, the flanges, the shaft and the box.
- Each geometry is solved in two modes:
 - anti: the rings at ± 1 , which gives the field at the null per volt across;
 - sym: both at +1, which gives the strays.
- Any supply is their superposition, exact for this linear problem [OC]: $E_{\text{null}} = k (V_B - V_A)$, with $k = 0.2546 \text{ (kV/cm)/kV}$ as built (0.2373 for the bands alone).
- The beads come from local solves with 0.025 mm cells, bounded by the hub's solution, and as drawn they also sit in the hub's maps (`sim/hub_beads_settled.py`).

The rings' field, as connected: $-15.0 / +15.0$ kV on the symmetric supply, 7.62 kV/cm and 2.57 Pa at the null, steady

Ring A on its own negative chain from the shaft (node 1), ring B on the positive one (node 4), two stages each; the bands on the glass from the polar to the equatorial bead. Finite volumes, 0.25 mm cells; the beads by local solves, 0.025 mm cells.



Ring A -15.0 kV, ring B $+15.0$ kV; 29.9 kV across, the null at the shaft's potential (0 V) by symmetry. Bands 26.25–55.71° from the axis, 29.9 mm apart along the glass (1.00 kV/mm). At the null 7.62 kV/cm, 2.57 Pa; within ± 5 mm it varies 4.4 % along the axis and 2.0 % across the equator. The supply's ripple moves it 0.014 kV/cm p-p (sim/hub_drift.py swing); the field does not swing. With the beads in their grooves as drawn (the bands alone give 7.10 kV/cm). Beads in the gel: polar 4.98, equatorial 4.46 kV/mm (design 5); in the glass 1.46 / 3.01. Each ring to the AH cores: 1.81 kV/mm on average. Settled (the insulators' leakage, 25 °C): 8.20 kV/cm at the null (sim/hub_drift.py).

Figure 8. The rings' field: $|E|$ in the section with equipotentials and field lines from B to A; the field and the pressure along the axis and across the equator; ring B's beads from local solves. Source: docs/figures/hub-rings-field.png (docs/make_rings_field_figure.py).

In space:

- **At the null:** 7.62 kV/cm along $-z$ (from B to A) and a pressure $\epsilon_0 E^2 / 2$ of 2.57 Pa, at switch-on. The beads add 7 % to the bands' 7.10: they are at the ring's potential and reach past its edges toward the gap.
- **Within ± 5 mm** it varies 4.4 % along the axis and 2.0 % across the equator.
- **Further out:** across the equator it rises to 8.5 kV/cm at r 16 mm. Along the axis it falls to zero at $|z| \approx 18$ mm and then turns toward the AH cores, because each band sits between the null and an AH core, both at 0 V.
- **In the insulation,** at switch-on:
 - 1.00 kV/mm along the glass between the rings, on average;
 - 4.98 / 4.46 kV/mm in the gel at the beads, as drawn;
 - 1.46 / 3.01 kV/mm in the glass under them;
 - each ring averages 1.81 kV/mm to the AH cores.
- **The retainer's ϵ** moves the field at the null by less than 2 %; the AH cores take about 13 % of it.

One revolution of the rotor: the pump goes through its phases 12 times, and the field at the null stays 7.62 kV/cm from B to A

sim/hub_revolution.py: the supply of record in ngspice (the symmetric pair, 2 + 2 stages, 100 pF, 100 GΩ per ring), settled over 280 cycles, then one revolution at 600 rpm each way (12 × 120 Hz cycles, 100 ms). The field at the null: $k(V_B - V_A)$, $k = 0.2546$ (kV/cm)/kV; the hub's insulators relax over minutes, so at 120 Hz the field follows the rings.

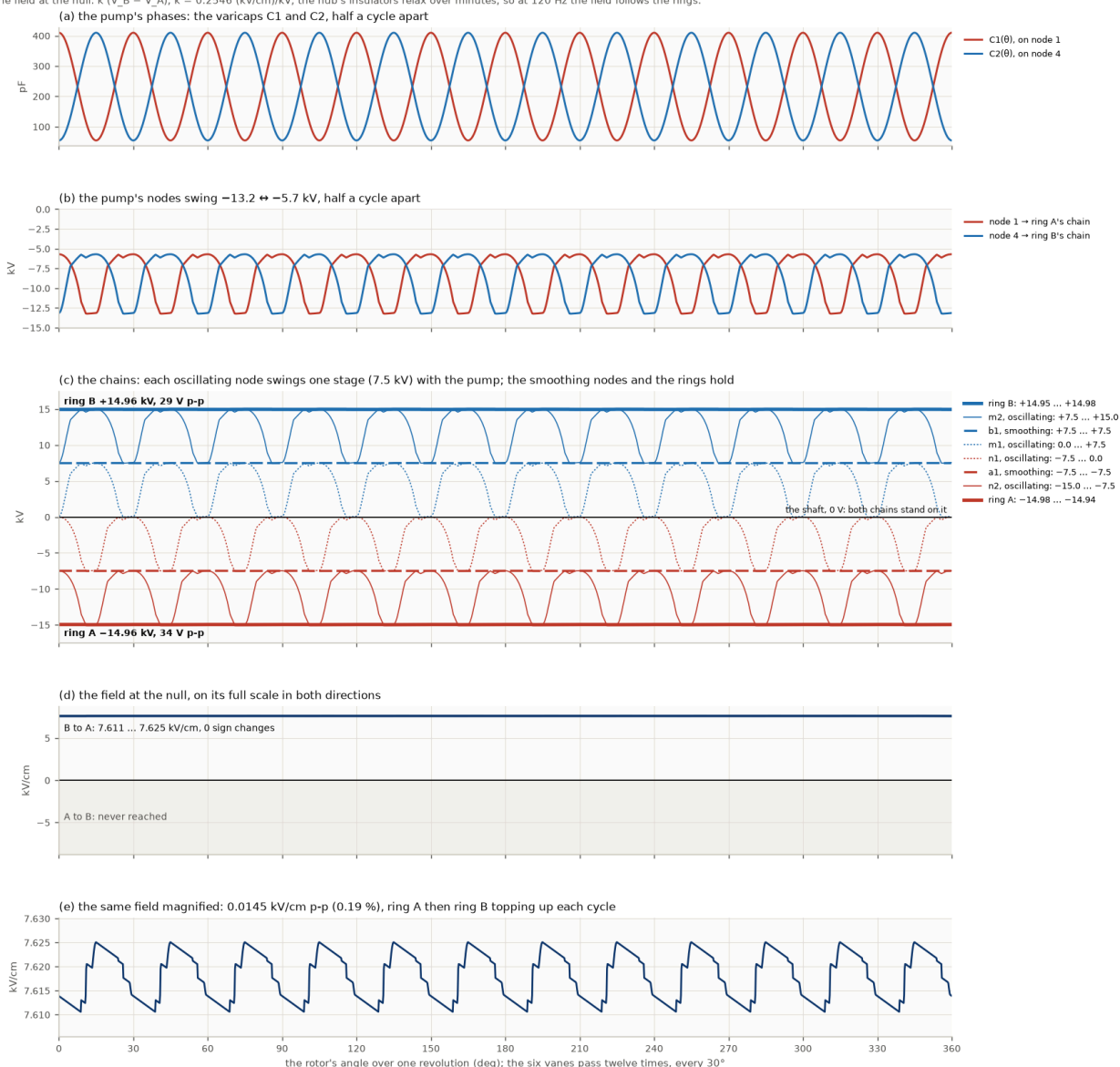


Figure 9. One revolution of the rotor: the pump goes through its phases twelve times (C1 / C2, nodes 1 / 4, the chains' oscillating nodes), while the rings hold and the field at the null stays 7.611–7.625 kV/cm from B to A. Source: docs/figures/hub-rings-revolution.png (sim/hub_revolution.py).

In time (the bench test's predictions are Figure 10, §5.3):

- **Start-up** from the pump's seed: 50 % at 0.12 s, 95 % at 0.24 s, 99 % at 0.33 s.
- **The ripple:**
 - ring A carries 34 V p-p and ring B 29 V p-p at 120 Hz;
 - ring A tops up as node 1 bottoms; ring B tops up 1.03 ms later, as node 4 peaks;
 - so the two mostly add: 57 V p-p across the gap, 0.014 kV/cm p-p (0.19 %) at the null.

- **No swing.** Over a whole revolution (12 pump cycles in 0.1 s) the field never changes sign.
 - Each chain's diodes pass charge one way only, and the storage capacitors hold the rings between cycles.
 - The hub turns with the rotor and the rings are symmetric about the shaft, so the rotation changes nothing at the null either.
 - A field swinging from A to B would need AC on the rings, through coupling capacitors. That gives about ± 1.8 kV/cm, with the pressure peaking near 0.14 Pa [IR estimate, not simulated with the rings].
- **The drift.** With the rings held at their DC, the insulators' leakage moves the potential from the electrostatic toward the conduction-settled state [OC]. The conductivities are datasheet-class [IR].
 - PEEK with gel at 25 °C: 7.62 → 7.95 (10 min) → 8.18 (1 h) → 8.20 kV/cm (6 h), +7.6 %, half-way at 7.6 min.
 - PEI gives +10.5 %, G10 +2.0 % (its σ/ϵ matches the glass's), and the glass at 40 °C +8.5 % in about a third of the time.
 - Settled, the beads relax and the PEEK takes the DC (§3.5).
- **The leakage:** the ledger of paths gives 158 G Ω per ring, against a 100 G Ω estimate. The chains' diodes and capacitors dominate; the rings' own insulation is about 200 T Ω (sim/hub-rings-build-findings.md §3).

3.8 The cost

Source: docs/cost/README.md. Every price is a placeholder for a one-off prototype, in EUR [RH].

The sheet (docs/cost/dccreg-air-build-cost-sheet.xlsx) costs all 2700 vane-matrix stacks with live formulas.

The placeholder targets:

- 6.5 kV/cm at the null, which needs 27.4 kV across the rings on the symmetric supply (on the bands' 0.2373: the rings as built give 7 % more, so the picks are conservative);
- at least 2 W, with $z \geq 1.3$;
- at most 6 + 6 vanes;
- corona-safe rims;
- 300 steady A-turns.

81 designs meet them.

build	design	power	total
cheapest that qualifies	6 mm gaps, 2.5 mm vanes, r 175 mm, 22°, 4 + 4	2.01 W	6,060
the stack of record (the designer kept it, 10-09)	6 mm gaps, 3 mm vanes, r 150 mm, 22°, 6 + 6	2.14 W	6,477
lowest cost per watt	8 mm gaps, 4 mm vanes, r 300 mm, 26°, 6 + 6	16.4 W	8,368 (510 per W)

The fixed parts are 3,754 before the 15 % contingency, about 70 % of the cheapest build (the BOM corrected at the settlement: one 22 mF part per coil, six spiders, the utron's SiFe as built):

- magnetic pump 1,552;
- mechanics 1,282, including the gear 250, the drive 250 and the frame 300;
- hub 626;
- electrostatic 180;
- AH 114.

The drive's first cut costs 1,066 against the sheet's 800 for the gear, the drive and the frame (docs/drive-gear-belt.md). The sheet keeps its placeholders until the designer accepts it.

The stacks' cost is per-part work, not metal (metal is about 8 %). Aluminium is the choice.

The symmetric supply costs 50 more than the asymmetric one: three more diodes and three more capacitors.

Not costed: re-sizing the AH or the pumps, the shaft for larger radii, design, test equipment, the vacuum system and tooling.

4. Fact sheet: the numbers of record, with their sources

Operating point and machine

item	value	source
speed	600 rpm each way, 1200 rpm relative	sim/pole-design-findings.md §6
pump frequency	120 Hz (6 sectors and 6 bridges per relative revolution)	sim/core_field.py; sim/pole_design_variants_op.json
tube length	940 mm, the record in solids (153 solids, no clash)	sim/tube-shaft-findings.md §0; sim/tube_geometry_record_results.json
masses	rotor 33.4 kg, counter-rotor 33.1 kg from the solids; the other studies' own counts give 32.9–35.3 and 33.4–34.9 kg	sim/shaft_bearings_record_results.json; sim/rotor-mechanics-findings.md §6; docs/drive-gear-belt.md §3
bearings	6 × 6205-class (25 / 52 / 15 mm) on 8 mm G10 spiders; 4 inner, 2 end; the counter-rotor floats on the 4 inner	sim/stack_sizing.py; sim/tube-shaft-findings.md §0
the shaft's modes	as laid out, d 25: f1 27–29 Hz, 123 μm gap change under 1 g lateral; d 40: 65 Hz, 25 μm; the eighth bearing pair: 131–165 Hz, 2–3 μm	sim/tube-shaft-findings.md §0; sim/rotor-mechanics-findings.md §4
shaft	d 25 (tube model) or d 30 (hub register), OPEN; non-magnetic, $\mu_r \leq 1.05$	sim/stack_sizing.py; presets/hub-locked.json; sim/ah-null-findings.md §5
the gap budget	total runout per inner bearing ≤ 0.027 mm as laid out (P5, axial spring preload), 0.05 mm with the eighth pair	sim/rotor-mechanics-findings.md §5
rotating loads	no banding (factors ≥ 19 at 750 rpm); balance G2.5 per body; windage about 12 W, bearings 5 W	sim/rotor-mechanics-findings.md
reference link	one inner bearing; 0.41 mA rms AC, no DC	sim/core_field_results.json
the drive (PROPOSED)	bevel reverser at side A, POM-C pinions; HTD 5M 4 : 1, 200 W motor; 0.48 / 0.52 N·m on the bodies, 1.00 into the frame; run-up 32 s	docs/drive-gear-belt.md

Magnetic pump (per side unless stated; sim/pole-design-findings.md §7–§8, sim/pole_design_variants_op.json)

item	value
utrons	3 at 0 / 120 / 240° on two 12 mm G10 carrier discs; r 57–130 mm; tips 14 mm; slot 30 × 30 mm; stack 100 mm of M235-35A 0.35 mm
coil	200 turns of 1.89 mm ² (Ø1.55 mm), 50 % fill, mean turn 319 mm; R 0.58 Ω; L 81.0 / 9.4 mH (κ 8.6); τ 0.140 s
neck (clamp)	80 % NiFe, 3.0 × 100 mm; Ψs 0.134 Wb-turns per group
bridges	6 M235-35A sectors, r 130.5–144.5 mm, 25.5° face, 0.65 kg each, in a G10 ring r 131.5–156.5 mm; side B offset 30°
gap	0.500 mm aligned, 9.74 mm unaligned; 0 clashes in 457 pairs
group	3 utrons in series: 0.243 H; 0.87–2.81 A per branch; 101 V peak
La / Lb	0.146 H DC chokes, 0.29 Ω, 0.87–1.15 A, ≤ 110 V (not designed)
D1*–D4*	Si, 0.55 V at 1 A, 2.8 A peak; reverse 112 / 101 / 43 / 45 V
gain	z 1.21 (z _{lin} , the screen's linear model); 1.139 loaded early (z _{early}), 1.147 with the bypass
kick	about 20 % of Ψs, once, at start-up: 0.110 A, 3.5 mJ seeded; above about 200 rpm
mass	2.23 kg per utron as built (SiFe 0.89, NiFe 0.15, Cu 1.07, G10 0.12)
power	belt 17.6 W (18.1 W with the bypass) + iron 1.25 W; coils 46 °C in air at a 40 °C ambient

AH pair (presets/hub-locked.json; sim/ah-steady-cusp-findings.md)

item	value
cores	Fair-Rite 4077484611, 77 MnZn, Ø12.3 × 41.3 mm at ±30.7–72 mm; G-10 formers ID 13.5 / OD 16.5 mm

item	value
coils	160 turns of 0.80 mm grade-1 wire in 4 layers, r 8.25–11.45 mm, ± 31.35 –71.35 mm; 1.01 mH; 0.333 Ω as wound (0.27 Ω in the deck); 1.22 W each, 43 °C
winding	two identical coils, start to the group's utrons, finish to the diode side; rotated copies, start leads toward the vessel
bypass (PROPOSED)	22 mF per coil, ESR 10 m Ω [RH]; LC 34 Hz; 0.65 A rms, about 0.5 V, 8 mW for the pair
field	290–308 A-turns per coil (300 mean, ± 3 %; 294 as wound); top and bottom within 17 A-turns
the null	at the centre; 0.113 T/m (0.111 as wound); 25 μ m per A-turn of imbalance; ± 0.44 mm over a cycle with the bypass, ± 6.4 mm without (sim/ah-null-findings.md)
rods	0.121 T at 300 A-turns; 0.30 T [RH] at 743 A-turns for the AH alone
77 MnZn	μ i 2000; 0.51 T at 400 A/m (25 °C); Br 0.18 T; Hc 20 A/m; 1 Ω -m; Curie above 200 °C (Fair-Rite's data sheet, as a search rendered it; to verify, presets/hub-locked.json AH core)

Electrostatic pump (per side unless stated; [sim/air-stack-sizing-findings.md](#) §6.5, [sim/air_stack_sizing_results.json](#), [sim/core_field_results.json](#))

item	value
vanes	6 stator (REF) + 6 rotor (node 1 / 4), Al 3 mm, full rounds R1.5; 11 gaps of 6 mm air
sectors	6 on a 60° pitch, 22° stator / 22° rotor; r 50–150 mm; 8° clearance each side at minimum C
rings of the vanes	stator ring r 150–162 mm into the G10 cage (r 162–166); rotor ring r 20.5–50 mm on the G10 sleeve (r 12.5–20.5)
C1 = C2	54.9–409.9 pF, κ 7.47 (4.81–37.3 pF per gap)
Ca = Cb	450.9 pF = 1.1 C _{max} : 6 full-annulus plates, 5 gaps of 6 mm, on the rotor
stack length	102 mm of C1 + 6 mm + 48 mm of Ca = 156 mm
operating peak	13.13 kV (19.7 kV breakdown / 1.5 [RH]); face field 21.9 kV/cm; rim 44.4 kV/cm (0.80 of onset)
gain	z 1.31 bare (1.3095 ngspice); 1.191 at start-up with the rings' chains
nodes	1 / 4: $-13.2 \leftrightarrow -5.7$ kV (mean -8.6); 2 / 3: $0 \leftrightarrow -6.1$ kV
clamps	Z1 / Z4 avalanche strings, BV 13.1 kV (66 $\times$ 200 V); 1.07 W, 0.95 mA peak each
D1–D4	reverse peaks 6.1 / 6.1 / 13.2 / 13.2 kV (parts not chosen)
power	belt 2.14 W, all into the clamps
aluminium (both sides)	rotor vanes 2.9 kg + Ca / Cb plates 6.1 kg; stator vanes 3.4 kg

Rings, supply and field ([docs/rings-design.md](#); [sim/hub_rings_build_results.json](#); [sim/hub_drift_results.json](#); [sim/hub_revolution_results.json](#))

item	value
bands	26.25–55.71°; 12.85 mm along the glass, 13.1 cm ² each; gap 29.92 mm; Cu-ETP 0.10 mm, 12 gores per band
beads	Ø3 mm polar (contact r 11.06, z ± 22.42 mm), Ø2 mm equatorial (contact r 20.65, z ± 14.08 mm)
ratings used	1 kV/mm along the glass; 5 kV/mm in the gel at a bead [RH]; 2 / 8 to qualify
supply	2 + 2 CW stages from the shaft; ring A -14.96 kV, ring B $+14.96$ kV; 8 diodes at 7.5 kV; 8 $\times$ 100 pF / 30 kV
k	0.2546 (kV/cm)/kV as built; 0.2373 for the bands alone
at the null	7.62 kV/cm, 2.57 Pa at switch-on (the bands alone 7.10); 8.20 kV/cm, 2.98 Pa settled at 25 °C; ± 5 mm uniformity 4.4 % (axis) / 2.0 % (equator)
fields in the insulation	glass along the gap 1.00 kV/mm on average; gel at the beads 4.98 / 4.46 kV/mm as drawn; glass at the beads 1.46 / 3.01 kV/mm; to the AH 1.81 kV/mm
settled	gel at the beads 0.69 / 1.88 kV/mm; the PEEK 2.3 kV/mm across the seat; the equatorial beads hold while $\sigma_{\text{glass}} \leq 4.1 \sigma_{\text{gel}}$
strays	5.1 pF to REF each, 1.1 pF between
time	95 % in 0.24 s; ripple 0.014 kV/cm p-p; one revolution 7.611–7.625 kV/cm; drift to 8.20 kV/cm in 6 h (PEEK, gel, 25 °C)
the hub's heat	coils 43 °C, glass 32–37 °C (32.5 °C between the rings) in a 25 °C room; the vessel 4.2e12 Ω -m there
leakage	100 G Ω per ring used (ledger 158 G Ω); 4.5 mW

What more stages would give (if the bench qualifies higher ratings;

sim/hub_rings_build_results.json best_by_family)

ratings (interface / gel)	stages a side	across	at the null (the bands alone)
1 / 5 kV/mm (the record)	2	29.9 kV	7.10 kV/cm (7.62 as built)
1 / 8	3	44.3 kV	7.80 kV/cm
2 / 5	2 (wider bands)	29.9 kV	8.35 kV/cm
2 / 8	4	57.9 kV	13.38 kV/cm

5. Status: decided, proposed, open

5.1 The record's parts by status

part	status	note
the 1 : −1 gear, the belt, two pumps per side	DECIDED	the gear and belt: a first cut, PROPOSED (docs/drive-gear-belt.md)
air for the first tests; full-round vanes; the 6 + 6 cap	DECIDED	
6 mm gaps, V _{op} = breakdown / 1.5; 22° / 22°; r 150; Ca = 1.1 C _{max}	DECIDED (10-09)	the designer kept the stack of record; the cost optimum differs (\$3.8)
the clamps Z1 / Z4	DECIDED (10-09)	
the HV side on the rotor; the link through one inner bearing	DECIDED	"for now"; a brush later
the magnetic pick (g 0.5, 6 bridges, 1200 rpm, 200 turns)	DECIDED (10-09)	the basis of DCCREG-UTR-101 (Rev A draft)
the steady cusp (22 mF bypass)	PROPOSED	drawn dashed on the schematic; it holds the null within ±0.44 mm
La / Lb, D1*-D4*, D1-D4, the clamp strings, the kick source	PROPOSED (first cuts)	sim/parts-first-cut-findings.md
the hub: layer order, sphere, wall, AH placement, symmetric pumps	DECIDED	
the rings outside the glass, copper foil, the symmetric supply	DECIDED	
the bands and beads	DESIGN	from the [RH] ratings; as drawn the polar bead sits 0.4 % under its rating
the PEEK retainer, the gel, the G10 coupler	DECIDED (materials, 10-09)	
the grooves' full-round tops, the AH ends rounded, the coupler as a tube, the leads' path	PROPOSED	docs/rings-design.md §2
the retainer split at the equatorial plane	PROPOSED	
the shaft's diameter and the bearings	OPEN	the designer's choice (\$5.2)
the ratings, the leakage, the conductivities	OPEN / ESTIMATE	the bench qualifies them

5.2 Open items

For the designer to decide (each has its numbers in this ledger):

- **the shaft and its bearings** (§3.1): six bearings with d 40 (65 Hz, the runout at 0.027 mm), or an eighth pair between the shaft and each bridge ring's outer end at d 25 (131–165 Hz, the runout at 0.05 mm; the layout +10–20 mm a side; stiff seats to keep it clear of 120 Hz);
- **the drive's first cut** (§3.1, docs/drive-gear-belt.md): accept it, or set the gear and belt otherwise;
- **the bypass and the split**, both PROPOSED;
- **the hub's first cuts** (§3.5): the grooves' full-round tops, the AH ends rounded to 1.5 mm, the coupler as a tube with the flanges turned to Ø60, the leads' path;
- **the AH's steady field**: 300 A-turns (294 as wound) against the 449 the AH was sized to; a steady 450 needs 240 turns or a larger pump;

- **the parts' first cuts** (sim/parts-first-cut-findings.md): La / Lb, the diodes, the clamp strings, the kick source and its polarity;
- **the fields' purpose:** what the field at the null is for is not documented (§2.1).

The bench (docs/bench-test-rings.md):

- the hold-off ratings (phase 1), which set the stages;
- each ring's leakage (phase 2);
- the conductivities (phase 3), at 25 °C and 40 °C: the equatorial beads need $\sigma_{\text{glass}} \leq 4 \sigma_{\text{gel}}$ at the hub's temperature;
- the field on the pump's own supply (phase 4).

Still open in the models and parts:

- the model checks of §6 (items 31–34): the neck, real diodes, the tube's strays and the utrons' κ in 3-D;
- the 77 MnZn data sheet: its values are in presets/hub-locked.json (AH core), as a web search rendered the sheet; re-read them from the current sheet before ordering;
- the capacitor mounts' creepage, potting and balancing: first cuts in sim/parts-first-cut-findings.md and sim/rotor-mechanics-findings.md §6;
- the gores' overlaps and joints, the fired-on coating's edges.

5.3 The bench test

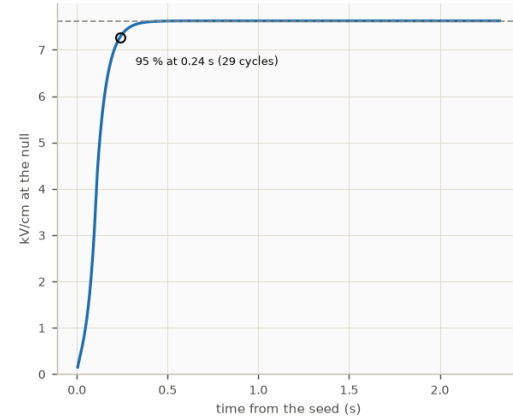
Source: docs/bench-test-rings.md.

- **The test hub:** the sphere with a Ø8 mm pumping tube at one pole ($\leq 1\text{e-}3$ Pa) and aluminium AH dummies at REF.
- **The instruments:**
 - a BGO electro-optic probe on a fibre along the axis;
 - ± 30 kV supplies through 100 M Ω ;
 - electrometers;
 - a DC partial-discharge detector at 10 pC.
- **The phases:**
 1. **Hold-off on coupons and the sphere:** ± 15.0 kV for 1 h, then ± 18.7 kV for 1 h, under 10 pC. This sets the ratings and the stages.
 2. **Leakage:** at least 100 G Ω expected.
 3. **DC drift on lab supplies for 6 h:** $7.62 \rightarrow 8.20$ kV/cm expected, with the return after grounding and a 40 °C repeat. The ratio of the glass's conductivity to the gel's, at the hub's temperature, decides the equatorial beads once settled.
 4. **The pump's own supply at 1200 rpm relative:** 95 % in 0.24 s and 0.014 kV/cm ripple expected; then the AH powered, with the Faraday offset measured first.

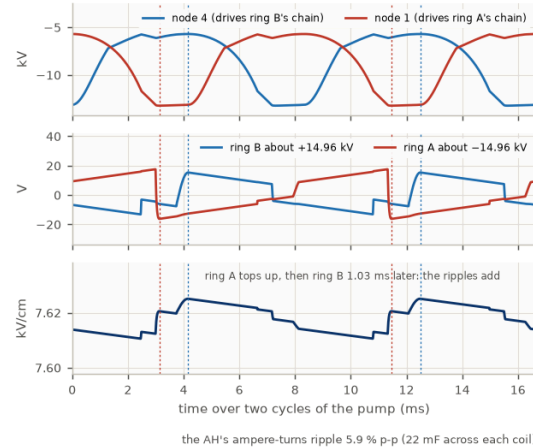
The bench test of the rings: what the field at the null should do (the symmetric supply, 2 + 2 stages, 29.9 kV across, bands 26.2–55.7°)

sim/hub_drift.py: the pump's supply in ngspice (100 pF storage, 100 GΩ per ring), the hub's finite volumes for the drift ($C \, dV/dt + G \, V = b$; conductivities datasheet-class [IR]). The field at the null as connected follows the DC across the rings; the leaking insulators then move it.

(a) the start-up: the pump self-excites and charges the rings through the multiplier (the lowest field each cycle)



(b) the 120 Hz ripple: 0.014 kV/cm p-p on 7.62 (0.2 %), no swing from A to B



(d) the bench test (docs/bench-test-rings.md)

1. hold-off

how: coupons, then the sphere: ramp the DC to flashover / breakdown; partial discharge at the design voltage

expect: sets the ratings: 1 kV/mm along the glass and 5 kV/mm in the gel (the record), 2 / 8 to qualify more stages

2. leakage

how: each ring from a lab supply; the current to REF on an electrometer; then the multiplier's parts

expect: $\geq 100 \, \text{G}\Omega$ per ring (the ledger: 158 GΩ); the diodes and capacitors dominate

3. DC drift

how: rings on lab supplies at the record's DC; the field at the null on the probe for 6 h

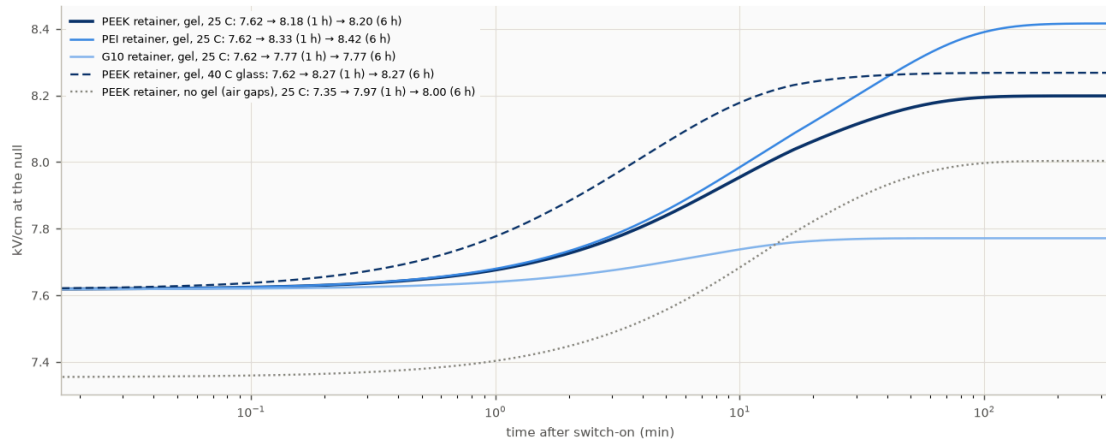
expect: 7.62 kV/cm at switch-on, 7.95 at 10 min, 8.18 at 1 h, 8.20 at 6 h (PEEK, gel, 25 °C)

4. the pump

how: the rings on the rotor's supply: start-up, ripple, then the drift again

expect: 95 % in 0.24 s; 0.014 kV/cm p-p at 120 Hz on 7.62: a steady field from B to A, no swing

(c) the drift: the field at the null after switch-on, the rings held at their DC (kV/cm: at switch-on → 1 h → 6 h)



The probe: an electro-optic (Pockels) BGO sensor on a fibre, entering through the pumping tube at one pole along the axis, where it barely disturbs the axial field. The bench's vessel must be evacuated: air inside would carry its own ions and screen the DC in minutes. The drift is the leakage redistributing the potential along the glass and through the gel and retainer: the field moves from the electrostatic value at switch-on toward the conduction-settled value. Fit $E(t)$ with one or two exponentials; the time constants check the conductivities, the end value their ratios.

Figure 10. What the bench test should see: the start-up, the 120 Hz ripple (the pump's nodes, the rings, the field), the 6 h drift by material, and the test's phases. Source: docs/figures/hub-bench-predictions.png (docs/make_hub_bench_figure.py).

6. Known inconsistencies and stale records

The lock recorded 18 and fixed none. The settlement (2026-10-09) resolved them against the baseline and found twelve more; each line says what was done and where. The model checks (31–34) run as studies of their own.

The lock's 18

#	what	resolution
1	the 3-D model and layout still carried the old vacuum 8 + 8 stack, Ca / Cb on the counter-rotor and the 120 mm placeholder hub	done: the record in solids, <code>sim/tube_geometry.py --record</code> → <code>docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.*</code> (940 mm, 153 solids, no clash); <code>sim/stack_sizing.py</code> puts the HV on the rotor
2	the shaft: d 25 in the tube, d 30 in the hub's register	analysed, the designer's: two bodies on the shaft (§3.1, §5.2)
3	<code>docs/figures/hub-locked.png</code> and its findings named the lock-down's rings as the record	done: retitled as the lock-down study; banner
4	<code>sim/core-field-findings.md</code> decisions 5 / 7, <code>sim/core-null-field-findings.md</code> called superseded designs "the record"	done: banners; "was the design of record"
5	the rotor schematic drew no bypass and quoted only the unbypassed A-turns	done: the bypass dashed (PROPOSED), both A-turn ranges, the kick box, the docstring
6	the cost sheet's BOM: 2 × 10 mF, the screen's SiFe mass, four spiders	done: one 22 mF part per coil, 0.89 kg, six spiders; every total +60
7	the kick: 38 mJ reported against about 3.5 mJ seeded; two labels	done: <code>sim/pole_design.py</code> <code>kick_seed</code> reports the seeded energy, 3.48 mJ at 0.110 A, on one basis
8	A / B naming; which AH coil in which branch; the winding sense	done: side A below in the record (naming notes); the winding rule (<code>sim/ah_winding.py</code>)
9	the magnetic gain on two bases	done: named, <code>z_lin</code> (the screen, 1.208) and <code>z_early</code> (the loaded run, 1.139; 1.147 with the bypass)
10	<code>record_supply</code> <code>E_pk_kV_cm</code> read as the null's field	done: <code>E_null_mean / pk / pp</code> added with the convention stated
11	the stack of record against the cost optimum	done: the designer kept the stack of record (§2.3)
12	the $\eta \approx 0.45\text{--}0.50$ read as the current machine's	done: <code>CONVENTIONS.md</code> and <code>README.md</code> place it with the disc machine
13	<code>README.md</code> , <code>CLAUDE.md</code> framed the Bennet browser tool	done: rewritten for the locked machine
14	<code>CONVENTIONS.md</code> switch naming: "the resonator rail"	done: REF, the shaft
15	<code>index.html</code> 's banner reads $\eta \approx 0.70$	frozen: flagged, not edited
16	<code>docs/commutator-design.md</code> , <code>docs/kicad/gap-topology-of-record.md</code> read as current	done: superseded banners
17	<code>CHANGELOG.md</code> undated; no design-loop or netlist v2–v4 entries	done: a dated timeline from the git history; the missing entries
18	some hub documents' tags under a redefinition	done: the hub documents and <code>presets/hub-locked.json</code>

Found and settled since

#	what	resolution
19	the frozen list named <code>charge-pump-synth-live.html</code> and <code>schematic.svg</code> at the root; they live in <code>tools/</code>	done: <code>README.md</code> , <code>CLAUDE.md</code> and §8; the check prints 0
20	the field at the null was solved for the bands alone	corrected: with the beads, 7.62 kV/cm at switch-on and 8.20 settled (§3.7); <code>sim/hub_rings_build.py --as-built</code> , <code>sim/hub_drift.py</code> and the figures carry it
21	the beads' fields came from a box filled with gel; the drawing's grooves bring the PEEK within 0.5 mm	corrected: 4.98 / 4.46 kV/mm as drawn (<code>sim/hub-beads-settled-findings.md</code> §3)
22	the AH coil: 160 turns in variant (a)'s window, its R scaled as bare copper	settled: 0.80 mm wire in 4 layers, 0.333 Ω ; the deck keeps 0.27 Ω (<code>sim/hub-thermal-findings.md</code> §1)

#	what	resolution
23	the AH rod's "about 600 A-turns" had the cone windings in series	corrected: 743 for the AH alone; the code's constant stays, conservative (sim/ah-null-findings.md §4)
24	"all the belt's power, about 21 W" missed the windage and the bearings	corrected: about 39 W in all (sim/rotor-mechanics-findings.md §5)
25	"the coils need banding"	corrected: no banding; preloaded studs and bonded joints (sim/rotor-mechanics-findings.md §1)
26	the runout of "about 0.05 mm or better"	corrected: 0.027 mm total per inner bearing as laid out, 0.05 mm with the eighth pair (§3.1)
27	the air-stack study took the shaft's first mode as above 30 Hz; the earlier grounded model gave 239 Hz	corrected: 27–29 Hz with the counter-rotor floating (§3.1)
28	"every reaction torque goes into the gear"	corrected: the gear puts the same torque on both bodies; the frame takes twice it (§1)
29	the cost sheet's six bearings; the drive makes seven, the eighth pair eight	open: with the shaft and the drive (§5.2)
30	the hub's register says d 30, but a 6205 takes a 25 mm journal	open: with the shaft (§5.2)

The model checks (running when this revision was written; their findings land in `sim/`)

#	check	where it lands
31	the neck, a nonlinear field check	sim/neck-nonlinear-findings.md
32	real diodes at start-up	sim/diodes-real-findings.md
33	the tube's strays, by a field solve	sim/tube-strays-findings.md
34	the utrons' κ in 3-D	sim/utron-3d-findings.md

7. Drawing register

The bundle, `docs/ledger/DCCREG-drawings-bundle.pdf`, carries every sheet below on A3, in this order. It opens with a two-page register and has bookmarks by part. The drawings stay in the repository at these paths, with the scripts that redraw them (`docs/ledger/register.py`).

Part A — the design of record (locked)

sheet	title	file
1	The locked machine: what drives what	docs/ledger/figures/architecture.png
2	Rotor circuits: reluctance pump and electrostatic pump	docs/schematic-rotor-circuits.svg
3	The rings' DC supply, the symmetric pair	docs/schematic-rings-supply.svg
4	DCCREG-UTR-101: the wound utron's SiFe half-core	docs/drawings/DCCREG-UTR-101.pdf
5	The wound utron against a bridge	docs/figures/utron-core-detail.png
6	The air vane stack, 6 mm gaps, 6 + 6 vanes	docs/figures/air-vane-stack-6mm.png
7	DCCREG-HUB-201: rings A and B, copper, beaded	docs/drawings/DCCREG-HUB-201.pdf
8	The rings' field at the null	docs/figures/hub-rings-field.png
9	The machine of record: section through the shaft	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-section.png
10	The machine of record: quarter cutaway	...-hub50-3d-cutaway.png
11	The machine of record: half section	...-hub50-3d-half.png
12	The locked hub in solids	...-hub50-hub.png
13	Reluctance sections A and B, plan cuts	...-hub50-reluctance-plan.png
14	Utron A1 with bridge A1, exploded	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-exploded.png
15	Reluctance section A, quarter cut	...-n8-wound-g0p5-6br-3d-reluctance.png
16	Reluctance section A, plan at mid-stack	...-n8-wound-g0p5-6br-3d-plan.png

Part B — supporting drawings and figures of the record

sheet	title	file
17	The lock-down study of the hub	docs/figures/hub-locked.png
18	The rings as built: retainer, edges, stages	docs/figures/hub-rings-build.png
19	What the bench test should see	docs/figures/hub-bench-predictions.png
20	One revolution of the rotor	docs/figures/hub-rings-revolution.png
21	Utron size against gain: the pick	docs/figures/utron-size-vs-gain.png
22	The vane matrix: what the radius buys	docs/figures/vane-matrix-radius.png
23	The vane matrix: rim corona margins	docs/figures/vane-matrix-corona.png
24	The AH's null in the locked hub	docs/figures/ah-null.png
25	The reversing gear and the belt drive (PROPOSED)	docs/figures/drive-gear-concept.svg

Part C — earlier phases, superseded, kept for the record

sheets	what	files
26-28	the machine as modelled before the record: section, cutaway, half section	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-{section,3d-cutaway,3d-half}.png
29	the first two-pump hub drive	docs/schematic-hub-drive.svg
30-31	the first pole pick's flux	docs/figures/pole-pair-flux-g0p5.png, -g1p0.png
32-34	the core-field studies: the pair inside, AC-coupled rings, floating cones	docs/figures/core-null-field.png, core-rings.png, core-swing-waveforms.png
35-37	the earlier spark-gap tube build	docs/geometry/tube/tube-r150-n8-{section,reluctance-plan,clocking-plan}.png
38-41	the C-EM and diode-core schematics	docs/schematic-diode-core-{switchless,dcbus}.svg, docs/schematic-cem-{in-discharge-path,motor-placement}.svg
42-43	the disc machine's KiCad schematic and its simplification	docs/kicad/DCCREG_Turbine_circuit.svg, schematic_simplification.png
44-46	the disc machine's cross-section, boomerang cap, placed motor	tools/cross-section.svg, docs/boomerang-cap.png, docs/geometry/motor/motor-il2f-6563b90d-rc40.png
47-50	the disc machine's field cuts	docs/geometry/rt/slices/*.png

CAD and DXF files (not on sheets; tools/step-viewer/ renders the STEP files):

- **of the record:**
 - docs/drawings/DCCREG-UTR-101_half-core_A.step and _lamination.dxf;
 - the record in solids: docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.step, with its .glb and .parts.json.
- **earlier phases:**
 - the machine as modelled before the record, tube-r150-n8-wound-g0p5-6br.step;
 - the spark-gap tube tube-r150-n8.step;
 - the disc builds freeze-v010-CaCb, floor-56b6cb83, opt-c9ac780b, il2-ce1a9380, il2f-6563b90d (with its pump + motor);
 - the designer's C-EM source, the two disc DXF layouts (r0.15, r0.6) and the KiCad source.
- **Not drawn as manufacturing drawings:**
 - the bridge, the bridge ring, the carrier discs and cheeks, the NiFe strip and the coil as parts;
 - La / Lb (first cut in sim/parts-first-cut-findings.md);
 - the AH coil and core (in the record's solids);
 - the kick source;
 - the gear (its concept is sheet 25);
 - the retainer and the coupler (in the record's solids; first cut in docs/rings-design.md §2).

8. Files, tools and how to regenerate

The records:

- `presets/hub-locked.json` (the hub's spec);
- `sim/hub_rings_build_results.json` (record, its as-built field, `record_supply`, `best_by_family`);
- `sim/pole_design_variants_op.json` (the magnetic pick);
- `sim/air_stack_sizing_results.json` and `sim/core_field_results.json` (the electrostatic pump);
- `sim/ah_steady_cusp_results.json`, `sim/hub_drift_results.json` and `sim/hub_revolution_results.json`;
- `sim/tube_geometry_record_results.json` and `sim/shaft_bearings_record_results.json` (the record in solids, its shaft).

The findings, one per study:

- `sim/pole-design-findings.md`, `sim/ah-steady-cusp-findings.md`, `sim/rotor-parts-duty-findings.md`;
- `sim/air-stack-sizing-findings.md`, `sim/vane-matrix-findings.md`, `sim/core-field-findings.md`;
- `sim/hub-locked-findings.md`, `sim/hub-rings-build-findings.md`;
- the settlement's: `sim/hub-beads-settled-findings.md`, `sim/hub-thermal-findings.md`, `sim/ah-null-findings.md`, `sim/rotor-mechanics-findings.md`, `sim/tube-shaft-findings.md` §0, `sim/parts-first-cut-findings.md`, `docs/drive-gear-belt.md`, and the model checks of §6 (31-34);
- the design and test documents `docs/rings-design.md` and `docs/bench-test-rings.md`;
- the cost guide `docs/cost/README.md`.

Regenerate, in order:

- **the studies:**
 - `sim/hub_rings_build.py --procs 4` (the rings; its `--as-built` step alone sets the field with the beads);
 - `sim/hub_drift.py --procs 4`;
 - `sim/hub_revolution.py`;
 - `sim/hub_beads_settled.py` and `sim/hub_thermal.py` (the hub settled, its heat);
 - `sim/ah_null.py`, `sim/rotor_mechanics.py`, `sim/drive_sizing.py --figure`, `sim/parts_first_cut.py`;
 - `sim/tube_geometry.py --record` and `sim/shaft_bearings.py --record` (the record in solids, its shaft).
- **the drawings and figures:**
 - `docs/make_rings_drawing.py`;
 - `docs/make_rings_field_figure.py`;
 - `docs/make_rings_supply_schematic.py`;
 - `docs/make_schematic_rotor.py`;
 - `docs/make_hub_rings_build_figure.py`, `docs/make_hub_bench_figure.py` and `docs/make_hub_revolution_figure.py`;
 - `docs/make_core_drawing.py` and `docs/make_utron_drawing.py`;
 - `docs/make_air_vane_drawing.py`.
- **the cost:** `docs/make_cost_sheet.py`, then recalculate the workbook.
- **this ledger:** `docs/ledger/make_ledger_figures.py`, then `docs/ledger/make_ledger.py`.

Frozen, unchanged at the lock and since:

- `shuttle_core.py`, `reference/`, `sim/pump_engine.py`, `sim/pump_sizing.py`, `sim/pump_synth.py`, `spice/`, `index.html`, `tools/pump-calc*`, `tools/pump-synth*`, `tools/charge-pump-synth-live.html` and `tools/schematic.svg`;
- the diff against `b33baa2` is empty.

Appendix A. Names and symbols

name	meaning
REF, node 0	the shaft, the rotor's electrical reference; the counter-rotor's stator vanes join it through one inner bearing
nodes 1 / 4	the rotor vanes of side A / B: the electrostatic pump's pumping nodes, $-13.2 \leftrightarrow -5.7$ kV
nodes 2 / 3	the transfer nodes between Ca / Cb and the diodes
C1(θ), C2(θ)	the varicaps, vanes of side A / B against REF; 55–410 pF
Ca, Cb	the transfer capacitors (1–2, 3–4), 451 pF
D1–D4	the electrostatic doubler's diodes: D1 2→0, D2 3→0, D3 1→3, D4 4→2
Z1, Z4	the clamps on nodes 1 / 4, avalanche strings at 13.1 kV
A1–A3, B1–B3	the wound utrons of side A / B; the groups L1(θ) / L2(θ)
La, Lb; D1*–D4*; Lp2, Lp3	the magnetic doubler's chokes, diodes and wiring strays
AH	the anti-Helmholtz coil pair on the axis; the null, its centre; the cusp, its field
ring A, ring B	the copper bands on the vessel, A below (-15.0 kV), B above ($+15.0$ kV)
Co / Cs, Dc / Dp; Coa / Csa, Dca / Dpa	ring B's and ring A's Cockcroft-Walton parts
m1, m2, b1; n1, n2, a1	the chains' oscillating and smoothing nodes
z	gain per pump cycle (the growth of the stored charge or flux)
κ	the swing ratio $C_{\text{max}} / C_{\text{min}}$ or $L_{\text{max}} / L_{\text{min}}$
V _{op}	the electrostatic pump's operating peak, set by the clamps
k	the field at the null per kV across the rings, (kV/cm)/kV
g	a gap (never d, CONVENTIONS.md §2)
θ	the rotor angle (rotor in code; never ϕ , the potential)